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BITS: SOFTWARE DEVELOPMENT

CHAPTER 1. INTRODUCTION

1.1 Introduction

The memorization of the Holy Quran, known as Tahfiz, is a highly respected Islamic tradition that requires dedication, discipline, and consistent practice. Traditionally, tahfiz students (known as tahfiz students) undergo tasmi' sessions where they recite memorized verses to their teachers, who then identify and correct errors in pronunciation (tajwid) and articulation points (makhraj). This one-on-one interaction is crucial for ensuring accurate recitation, as even minor mistakes can alter the meaning of Quranic verses.

However, the traditional tasmi' system faces several challenges in the modern educational context. Tahfiz institutions often have high student-to-teacher ratios, limiting the frequency and duration of individual tasmi' sessions. Students who wish to practice independently during self-revision (muraja'ah) lack immediate feedback on their recitation accuracy. Furthermore, teachers have limited tools to track student progress systematically over time, making it difficult to identify persistent error patterns or measure improvement.

Recent advancements in artificial intelligence, particularly in automatic speech recognition (ASR) and rule-based expert systems, present an opportunity to address these challenges. ASR technology can convert spoken recitation into text or phonetic representations, while expert systems can apply predefined Tajwid rules to detect pronunciation errors. By integrating these technologies into a mobile application with a teacher monitoring portal, it becomes possible to create an intelligent system that supports both independent student learning and teacher supervision.

This project, named TasmiqAI, aims to develop such an AI-based Quran recitation assessment and monitoring system for tahfiz students. The system will allow students to record their recitation of selected Quran verses through a mobile application. The recorded audio will be processed by an AI engine that performs speech recognition, applies Tajwid rule checking, and generates an accuracy score with detailed feedback on specific errors. Teachers will be able to access a web portal to review student recitations, monitor progress through analytics dashboards, and provide additional feedback.

This report documents the complete development process of TasmiqAI following the Waterfall software development methodology. The report covers the analysis of existing

systems and requirements, the design of system architecture and database, the implementation of all system components using the specified technologies, and the testing procedures and results.

1.2 Problem statement(s)

Based on observations of the current tahfiz education system and a review of existing technology solutions, several key problems have been identified that motivate this project:

1.2.1 Limited Access to Personalized Feedback During Self-Revision

- Tahfiz students often practice recitation independently during muraja'ah (revision) sessions outside of formal tasmi' hours. During these self-directed practice sessions, students have no immediate way to verify the accuracy of their recitation, particularly regarding subtle Tajwid rules such as Idgham (merging), Ikhfa (hiding), and Mad (prolongation). Errors practiced repeatedly become ingrained, requiring significant effort to correct later.

1.2.2 High Teacher Workload Limiting Individual Tasmi' Frequency

- In typical tahfiz institutions, each teacher is responsible for supervising 20 to 50 students. Conducting individual tasmi' sessions for every student daily is impractical. As a result, each student receives limited direct teacher feedback, typically once or twice per week. This constraint slows the learning progress and limits opportunities for error correction.

1.2.3 Lack of Systematic Progress Monitoring Tools

- Teachers currently rely on manual records, such as notebooks or spreadsheets, to track student progress. This approach is time-consuming, error-prone, and does not facilitate data-driven analysis of student performance. Teachers cannot easily identify class-wide error patterns, track individual improvement over time, or generate performance reports for parents or administration.

1.2.4 Inefficient Error Identification in Recitation

- Even when teachers conduct tasmi' sessions, the process of manually listening to each verse and identifying specific Tajwid errors is mentally demanding. Teachers must simultaneously listen to recitation, recall correct pronunciation rules, and document errors. This cognitive load can result in missed errors or incomplete feedback.

1.2.5 Limited Integration of Modern Technology in Quran Education

- While many educational domains have embraced technology-assisted learning (e.g., language learning apps, math tutoring systems), Quran recitation learning remains largely traditional. Existing Quran applications either focus on reading text without pronunciation analysis or provide limited audio comparison features without proper Tajwid rule checking.

1.3 Objective

This project aims to address the identified problems through the following specific objectives:

1.3.1 To develop an AI-based Quran recitation assessment system that processes student audio recordings and generates automated feedback on recitation accuracy using automatic speech recognition and expert system technologies.

1.3.2 To implement speech recognition and expert system rules to detect Tajwid and Makhraj errors in student recitations, providing specific identification of error types and locations within verses.

1.3.3 To design a mobile application that provides students with real-time feedback, accuracy scores, and progress tracking for independent muraja'ah practice.

1.3.4 To develop a web-based teacher monitoring portal that enables teachers to review student recitations, access performance analytics, provide feedback, and manage class activities.

1.3.5 To evaluate the system performance using accuracy metrics for the AI assessment component and usability testing methods for the overall system.

1.4 Scope

The scope of this project is defined across multiple dimensions :

1.4.1 Modules to be developed

Module	Description
User Authentication and Management	Handles student and teacher registration, login, password reset, and role-based access control (student vs. teacher).
Student Mobile Application	Allows students to select surahs, record recitations, view AI feedback, track personal memorization progress, and access recitation history.
Audio Recording and Processing	Captures audio through mobile device microphone, converts to appropriate format (WebM/MP3), and prepares for transmission to backend.
Speech Recognition Module	Converts student recitation audio into Arabic text transcription using pre-trained Hugging Face models (Whisper/Wav2Vec2).
Tajwid Expert System Module	Applies rule-based reasoning to detect Tajwid errors including incorrect makhraj, missing mad extension, incorrect Idgham/Ikhfa application, and verse sequence errors.
Recitation Evaluation Module	Calculates accuracy scores, identifies specific error locations, and generates human-readable feedback for students.
Teacher Web Portal	Provides dashboard with class overview, individual student progress views, recitation review interface, and feedback

	submission tools.
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1.4.2 Target Users

User Type	Description
Tahfiz Students	Primary users who will use the mobile application to practice recitation and receive AI feedback.
Tahfiz Teachers	Users who will access the web portal to monitor student progress, review recitations, and provide feedback.
Tahfiz Institutions	Organizations that may adopt the system for their tahfiz programs (administration features limited).

1.4.3 Quranic Content Scope

The TasmiqAI system has been developed to support complete Quranic coverage, encompassing all 114 surahs across the full 30 Juz.

While the initial development phase focused on selected surahs from Juz' Amma (Surah Al-Fatihah, An-Nas, Al-Falaq, Al-Ikhlās, and Al-Mulk verses 1-15) due to computational and dataset constraints, the final implementation successfully expanded to include the entire Quran. The scalable system architecture enabled this expansion without requiring significant redesign of core components

Students can now practice recitation of any verse from any surah, and teachers can monitor progress across the complete Quran curriculum.

1.4.4 Technical Scope Limitation

Aspect	Scope
Platform	Mobile: Android and iOS (via Expo Go cross-platform development); Web: Modern browsers (Chrome, Firefox, Safari, Edge)
Language Support	Arabic text recognition (AI engine processes Arabic recitation); User Interface (UI) available in English and Malay (bilingual)
Real-time Processing	Audio processed after upload completion (not live streaming or real-time during recitation)
Tajwid Rules	Major rules implemented: Makhraj (articulation points), Mad (prolongation), Idgham (merging), Ikhfa (hiding), Izhar (clear pronunciation), Iqlab (conversion)

1.5 Project Significance

The successful development of TasmiqAI will provide significant benefits to multiple stakeholders:

1.5.1 For Tahfiz Students:

- Enables independent practice with immediate, personalized feedback
- Reduces reliance on teacher availability for basic error detection

- Provides visual feedback highlighting specific error locations
- Tracks personal progress to motivate continued improvement
- Supports consistent muraja'ah through accessible technology

1.5.2 For Tahfiz Teachers:

- Reduces workload by automating basic error detection
- Provides data-driven insights into student and class performance
- Enables efficient review of student recitations through organized interface
- Facilitates identification of common error patterns for targeted instruction
- Generates automatic progress reports for parent-teacher meetings

1.5.3 For Tahfiz Institutions:

- Modernizes Quran education through technology integration
- Provides standardized assessment framework for recitation quality
- Enables systematic documentation of student progress
- Supports remote learning scenarios where in-person tasmi' is limited

1.5.4 For Islamic Education Research:

- Contributes a working implementation of AI for Quran recitation assessment
- Provides a framework for evaluating ASR performance on Quranic Arabic
- Demonstrates integration of expert systems with modern deep learning approaches

1.6 Expected Output

The expected outputs of this project are as follows:

1.6.1 AI-Based Quran Recitation Assessment Engine

- Python-based engine using Hugging Face Transformers for speech recognition
- Rule-based expert system for Tajwid rule checking
- Audio preprocessing pipeline using Librosa and FFmpeg
- Scoring algorithm producing accuracy percentages (0-100%)

1.6.2 Student Mobile Application (React Native + Expo)

- User authentication screen (login/register)
- Dashboard showing overall progress percentage and today's goal
- Recitation module with surah/verse selection and recording interface
- Feedback display showing score and specific error details
- Progress tracking with history view of past recitations
- Learning history with detailed session information

1.6.3 Teacher Web Portal (React + Vite + TailwindCSS)

- Dashboard with key metrics (total students, average score, new students)
- Student list and individual student detail view
- Recitation review interface with audio playback and AI assessment results
- Feedback submission form for teacher comments
- Analytics with performance charts and common error reports
- Class management for grouping students

1.6.4 Backend System (FastAPI + Supabase)

- RESTful API endpoints for all frontend operations
- User authentication and authorization (JWT-based)
- Audio file upload and storage management
- Integration with AI engine for recitation processing
- Database schema for users, recitations, assessments, classes, feedback

1.6.5 Documentation

- Complete project report covering all SDLC phases
- User manual for both student app and teacher portal

- Technical documentation for future maintenance and expansion

1.7 Conclusion

This chapter has introduced the TasmiqAI project, which aims to develop an AI-based Quran recitation assessment and monitoring system for tahfiz students. The background of traditional tahfiz education was presented, highlighting the challenges of limited teacher availability, lack of independent feedback during self-revision, and insufficient progress monitoring tools.

Five specific problem statements were identified, directly motivating the development of this system. Five objectives were established to address these problems, focusing on AI assessment development, Tajwid rule implementation, mobile application design, teacher portal development, and system evaluation.

The scope of the project was defined, covering the modules to be developed, target users, Quranic content limitations, and technical constraints. The project significance was described for students, teachers, institutions, and research communities. Finally, the expected outputs were listed, including the AI engine, mobile application, teacher portal, backend system, and documentation.

The next chapter will present a literature review of existing Quran learning systems and relevant technologies, followed by a detailed description of the project methodology and requirements.

CHAPTER 2. LITERATURE REVIEW AND PROJECT METHODOLOGY

2.1 Introduction

This chapter provides a comprehensive review of existing literature and technologies relevant to the development of the TasmiqAI system, an AI-based Quran recitation assessment and monitoring platform for tahfiz students. The chapter begins by identifying the primary domains that inform this project, namely Islamic education technology, automatic speech recognition for Arabic, and expert systems for rule-based error detection. Following this domain identification, existing Quran learning systems are examined and analyzed to identify gaps in current solutions that TasmiqAI aims to address. Alternative technical approaches are then discussed and justified to explain why the chosen methodologies and tools are most appropriate for this project. The Waterfall software development methodology is described as the framework guiding the project through its sequential phases. Finally, the project requirements including software, hardware, and other resources are listed, followed by the project schedule and key milestones. As noted by Creswell (2018), a thorough literature review serves not only to summarize existing knowledge but also to position the current research within the broader academic conversation and to justify the research approach taken.

2.2 Facts and findings (based on topic)

2.2.1 Domain

The TasmiqAI initiative functions at the confluence of three fundamental domains: Islamic educational technology, automated speech recognition for the Arabic language, and expert systems designed for rule-based error detection. Each of these domains

provides critical insights and methodologies that are integral to the formulation of an AI-driven system for assessing Quranic recitation.

The realm of Islamic educational technology has witnessed substantial advancement over the preceding decade. As elucidated by Salim et al. (2025), the applications and platforms dedicated to Quranic education have transitioned from rudimentary text displays to sophisticated interactive systems that encompass audio recitations, translations, and tools for memorization. Nevertheless, these authors emphasize that the deployment of artificial intelligence for the automated evaluation of recitation quality constitutes a nascent field with considerable potential to enhance equitable access to high-quality feedback.

The authors further assert that "the utilization of deep learning for the recognition of Tajweed laws signifies a pioneering advancement in Quranic educational technology" (Salim et al., 2025, p. 3).

In the field of automatic speech recognition (ASR) pertaining to the Arabic language, notable progress has been achieved with the emergence of deep learning models such as Whisper, developed by OpenAI, and Wav2Vec2, developed by Meta. These models have undergone pre-training on extensive multilingual datasets and can be fine-tuned to address specific domains, including Quranic recitation. Shaiakhmetov et al. (2025) elucidate that the assessment of Tajweed rules via deep neural networks necessitates models proficient in discerning subtle phonetic distinctions that possess significance within the Quranic framework but may be deemed acceptable in colloquial Arabic communication. This presents a distinct challenge for ASR systems employed in the evaluation of Quran recitation.

The third domain pertinent to TasmiqAI pertains to expert systems designed for rule-based error detection. Expert systems represent a category of artificial intelligence applications that replicate the cognitive decision-making processes of human specialists

through the application of a defined set of rules to established facts. In the context of Tajwid assessment, an expert system is capable of encoding the principles of Tajwid as conditional statements, typically articulated in the form of IF-THEN propositions, which assess a student's recitation in relation to the standards of correct pronunciation. This methodology serves as a complementary mechanism to deep learning-based Automatic Speech Recognition (ASR). As observed by Mosul University (2025), the ASR component delivers a phonetic representation of the student's recitation, whereas the expert system assesses that representation against the codified Tajwid principles to pinpoint particular inaccuracies.

2.2.2 Existing System

Several existing systems address Quran learning with varying degrees of technology integration. These systems range from commercial mobile applications to academic research prototypes, each with distinct features and limitations.

Tarteel (2024) is a commercially available AI-powered Quran memorization application available on both iOS and Android platforms. The application offers features including verse matching, progress tracking, and error detection by comparing student recitation to reference recitations. However, the proprietary algorithms used by Tarteel are not publicly disclosed, and the application operates on a subscription-based model. Furthermore, Tarteel does not provide a teacher monitoring portal, which limits its utility in formal tahfiz educational settings (Tarteel, 2024).

Another commercial application, Qara'a (2024), focuses on pronunciation learning for Quranic Arabic. The application provides word-by-word audio and tracks user progress through various learning modules. However, Qara'a offers limited Tajwid rule checking capabilities and primarily focuses on reading rather than memorization assessment. This makes it less suitable for tahfiz students who need to assess their memorized recitation (Qara'a, 2024).

TajweedMate (2023) is an Android application designed specifically for learning Tajwid

rules through interactive exercises and quizzes. While the application effectively teaches theoretical knowledge of Tajwid rules, it does not incorporate speech recognition technology. Consequently, students cannot practice actual recitation and receive automated feedback on their pronunciation (TajweedMate, 2023).

Several academic research efforts have also contributed to this domain. Salim et al. (2025) developed a research prototype called the Tahfiz Intelligent System, which includes student progress tracking and a teacher dashboard. However, this system has limited speech recognition capabilities and remains at the research prototype stage rather than a production-ready application.

Based on the review of existing systems, several gaps have been identified. First, there is limited integration of AI for detailed Tajwid checking. While applications like Tarteel use artificial intelligence for verse matching, specific identification of Mad duration errors or Makhraj inaccuracies is not explicitly documented in the available literature. Second, most existing applications focus exclusively on the student experience without providing teachers with tools to monitor class progress, review recitations, or provide structured feedback. Third, commercial applications do not disclose their artificial intelligence methodologies, making it difficult for academic research to build upon their work. Fourth, as noted by 'Abdelfattah et al. (2025), the availability of labeled Quranic recitation datasets with Tajwid annotations remains a significant challenge for research in this domain.

TasmiqAI distinguishes itself from existing systems through several key features. The system integrates automatic speech recognition with a rule-based expert system to provide detailed Tajwid checking. It provides separate interfaces for students (mobile application) and teachers (web portal) to enable complete monitoring capabilities. The system uses open-source technologies including FastAPI, Hugging Face Transformers, and Supabase, which enables transparency and reproducibility. Finally, the project maintains an academic focus with documented methodology and systematic evaluation.

2.2.3 Technique

Several alternative technical approaches were considered during the development of the TasmiqAI system. This section discusses these alternatives and justifies the selection of the chosen approaches for each system component.

For automatic speech recognition, three primary alternatives were evaluated. The first alternative was Google Cloud Speech-to-Text, a commercial API service with Arabic language support. While this service offers high accuracy, it requires an internet connection for every API request, incurs costs per API call, and provides limited control over the underlying model (Google Cloud, 2024). The second alternative was Vosk, an offline speech recognition toolkit. However, Vosk demonstrates lower accuracy for Arabic compared to commercial alternatives and has limited vocabulary coverage for Quranic terms (Vosk, 2023). The third alternative was developing a custom convolutional neural network from scratch trained on Quranic recitation data. This approach would require a massive labeled dataset of hundreds of hours of recitation audio, which was not available, and would be computationally expensive and time-consuming (Shaiakhmetov et al., 2025). Therefore, the chosen approach was to utilize Hugging Face Transformers with pre-trained Whisper or Wav2Vec2 models. These models provide good Arabic support out-of-the-box, are open-source and freely available, and can run locally or on a server with optional GPU acceleration.

For the backend framework, three alternatives were considered. Django is a full-featured Python web framework that includes many built-in components. However, Django is relatively heavyweight for an API-only application, uses synchronous request handling by default, and has a steeper learning curve (FastAPI Documentation, 2024). Flask is a lightweight Python web framework that offers simplicity and flexibility. However, Flask lacks built-in data validation, requires multiple extensions to add functionality, and offers slower performance for asynchronous operations compared to FastAPI. Node.js with Express was also considered as a JavaScript-based backend alternative. However, Node.js has inconsistent artificial intelligence library availability for Arabic speech recognition and would require a separate Python service to run the AI engine (FastAPI

Documentation, 2024). The chosen approach was FastAPI, which offers asynchronous request handling for better performance, automatic OpenAPI documentation generation, built-in data validation using Pydantic, and seamless integration with Python artificial intelligence libraries including PyTorch and Hugging Face Transformers.

For database management, three alternatives were evaluated. Firebase Firestore is a NoSQL document database that offers real-time synchronization. However, Firestore is less suitable for the relational data structure required for users, classes, and recitations, and it requires more complex queries for reporting and analytics (Supabase Documentation, 2024). Local MySQL running on XAMPP is a traditional self-hosted relational database. This approach requires manual setup of authentication, file storage, and API layers, which significantly increases development overhead. MongoDB is another NoSQL document database that offers flexible schema design. However, MongoDB is less ideal for teacher dashboards that require complex joins and aggregations across multiple data tables. The chosen approach was Supabase, which provides PostgreSQL reliability with built-in authentication, automatic REST API generation, real-time subscriptions, and row-level security. As noted in the Supabase documentation (2024), this significantly reduces backend development time compared to building these features from scratch.

For mobile application development, three alternatives were considered. Native Android development using Kotlin would provide optimal performance and full access to device features. However, this approach would require separate iOS development using Swift, effectively doubling the development time and effort for cross-platform deployment (React Native Documentation, 2024). Flutter, Google's cross-platform framework using the Dart language, offers good performance and a single codebase for both platforms. However, Flutter requires learning the Dart language, which has a smaller ecosystem compared to JavaScript, and there are fewer community-contributed plugins for audio recording compared to React Native. Ionic is a web-based hybrid framework that uses standard web technologies. However, Ionic has performance limitations for audio processing applications and provides a less native user experience compared to React

Native. The chosen approach was React Native with Expo, which enables cross-platform development for both Android and iOS from a single JavaScript codebase, has extensive community support with many available plugins, and provides robust audio recording capabilities through the Expo AV library (Expo Documentation, 2024).

For the teacher web portal, three alternatives were evaluated. Create React App (CRA) is the traditional React build tool that offers a standard development experience. However, CRA has a slower development server for large applications, produces larger production bundle sizes, and requires more manual configuration for optimization. Next.js is a React framework that offers server-side rendering and static site generation. However, Next.js is overly complex for the teacher portal requirements, and server-side rendering is not necessary for an authenticated dashboard application (Vite Documentation, 2024). Plain HTML, CSS, and JavaScript without any framework would be difficult to maintain as the application grows in complexity and would provide no component reusability. The chosen approach was React with Vite, which provides a fast development server with hot module replacement, optimized production builds with code splitting, and seamless TailwindCSS integration for rapid user interface development.

It is important to acknowledge a significant limitation of the current system: the AI speech recognition engine has limited accuracy for Quranic Arabic pronunciation. This limitation stems from several factors that are documented transparently in this report. First, Arabic contains phonemes that are inherently difficult to distinguish, even for human listeners, and these distinctions are particularly critical for proper Tajwid. ASR models trained on general Arabic speech rather than Quranic recitation specifically may not capture these subtle distinctions. Second, there is limited availability of labeled Quranic recitation datasets with Tajwid annotations that could be used to fine-tune the ASR models for this specific domain. As noted by 'Abdelfattah et al. (2025), the scarcity of such datasets remains a significant challenge for research in this area. Third, training or fine-tuning large ASR models requires significant GPU resources that were not available for this project. This limitation is acknowledged as the primary area for future improvement, rather than presenting the system as having perfect accuracy.

2.3 Project Methodology

This project follows the Waterfall Software Development Life Cycle methodology, which is a sequential, linear approach where each phase must be completed and reviewed before the next phase begins (Sommerville, 2016). The waterfall model is one of the earliest and most widely documented software development methodologies, and it remains appropriate for certain types of projects despite the subsequent development of agile methodologies. According to Pressman (2014), the waterfall model is most suitable for projects with well-understood requirements, clear scope boundaries, and sequential dependencies between phases where later phases cannot begin until earlier phases are complete.

The waterfall methodology is appropriate for the TasmiqAI project for several specific reasons that align with the characteristics Pressman (2014) identifies. First, the project requirements are relatively well-understood from the proposal phase, having been developed through discussions with the supervisor and review of existing systems, reducing the risk of major requirement changes during development. Second, the project has a defined scope with clear objectives and specific deliverables, which aligns with the waterfall model's emphasis on upfront planning and scope definition. Third, the system components including the mobile application, web portal, AI engine, and backend have clear interfaces that can be designed before implementation begins, allowing for sequential development. Fourth, the project duration is fixed within the academic calendar, with PSM 1 spanning approximately 14 weeks and PSM 2 another 14 weeks, requiring a predictable, plan-driven approach.

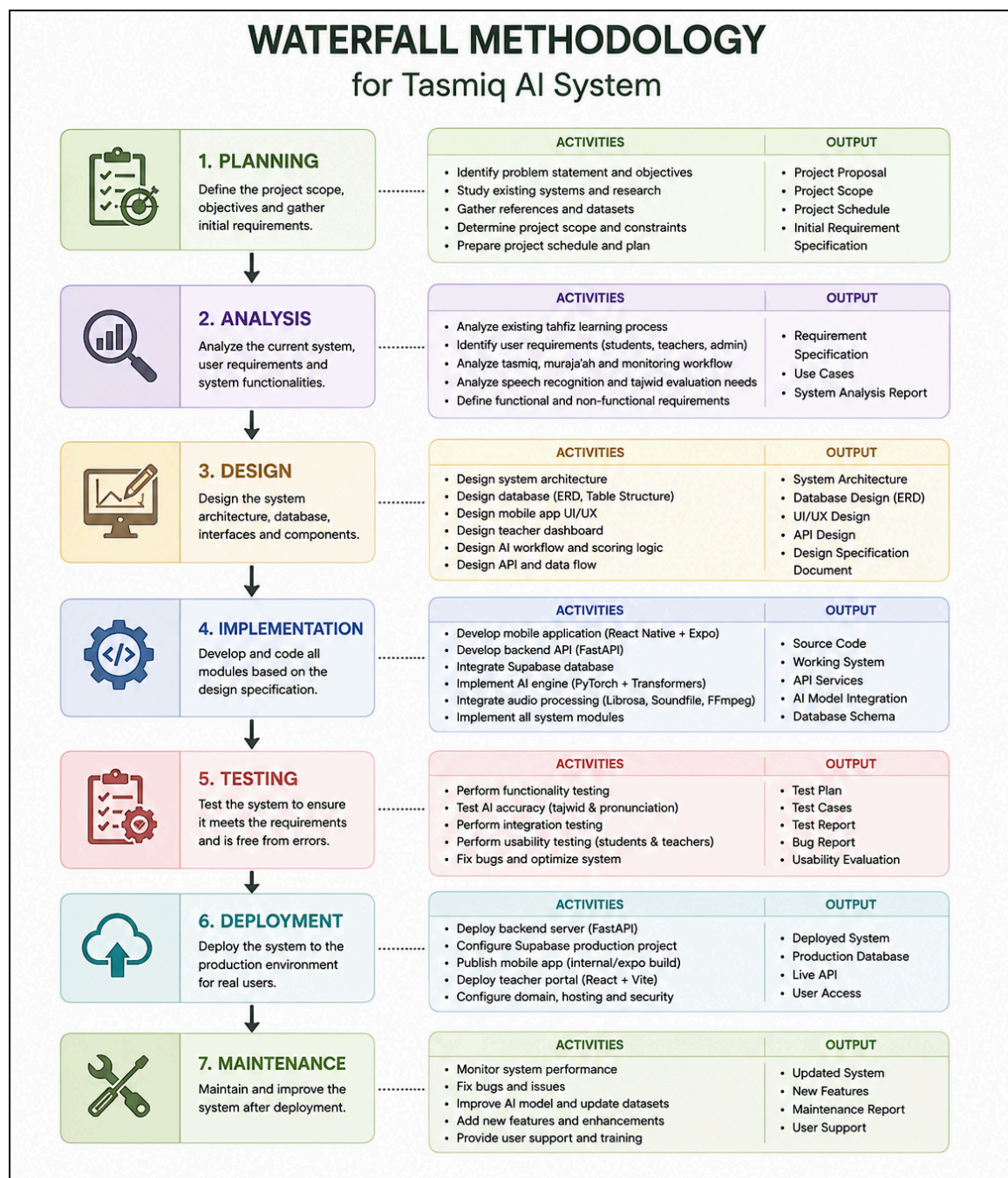


Figure 2.1 illustrates the sequential phases of the Waterfall SDLC methodology as applied to the TasmiiqAI project. The diagram shows the linear progression from planning through analysis, design, and implementation, followed by the verification and testing phases. The feedback arrows indicate that while the phases are sequential, issues discovered in later phases may require revisiting earlier phases.

a) Planning Phase

In the planning phase, which occupied the first four weeks of PSM 1, the project proposal was developed. This phase involved identifying the problem statements based on

observations of current tahfiz education practices, formulating specific objectives that the system would achieve, defining the scope boundaries including which modules would be developed and which surahs would be covered, and describing the expected outcomes. A project schedule was created with weekly milestones and final deliverables for both PSM 1 and PSM 2. Required resources including software tools, hardware specifications, and datasets were identified and documented. The proposal was submitted for supervisor approval and faculty review, and upon approval, the project proceeded to the analysis phase.

b) Analysis Phase

The analysis phase, which occupied weeks five through eight of PSM 1, involved investigating the current tahfiz system to understand student and teacher needs in depth. This phase included conducting informal interviews with tahfiz students and teachers to understand their challenges and requirements. Functional requirements were documented, specifying what the system must do, including recitation recording, AI assessment, feedback display, and teacher monitoring. Non-functional requirements were also documented, specifying how well the system should perform, including response time, accuracy expectations, and usability considerations. Data requirements were analyzed to design the database schema, identifying entities such as users, recitations, assessments, classes, and feedback. Use case diagrams and sequence diagrams were developed to represent system interactions and to validate that the requirements captured all necessary functionality. This phase is documented in detail in Chapter 3 of this report.

c) Design Phase

The design phase, which occupied weeks nine through twelve of PSM 1, translated the requirements developed in the analysis phase into concrete system specifications ready for implementation. High-level system architecture was defined showing component interactions between the mobile app, backend server, AI engine, and database. Database design was completed using entity-relationship diagrams and normalization to third normal form, ensuring data integrity and eliminating redundancy. API endpoints were designed for FastAPI, specifying request and response formats for each operation including user authentication, recitation submission, score retrieval, and teacher

dashboard queries. User interface wireframes and navigation flows were created for both the mobile application and the teacher web portal, defining screen layouts, navigation paths, and user interaction patterns. This phase is documented in detail in Chapter 4 of this report.

d) Implementation Phase

The implementation phase, which occupies weeks one through nine of PSM 2, involves coding all system components according to the design specifications developed in the previous phase. The FastAPI backend is developed with authentication endpoints, recitation upload handling, and API routes for both student and teacher functions. The AI engine is implemented using Hugging Face Transformers for speech recognition, Librosa for audio preprocessing including noise reduction and silence trimming, and a rule-based expert system for Tajwid rule checking. The React Native mobile application is developed with Expo, including screens for login, dashboard, surah selection, recitation recording, feedback display, and history viewing. The React teacher portal is developed with Vite and TailwindCSS, including dashboard views, student lists, recitation review interfaces, and analytics charts. Database tables are created in Supabase with appropriate indexes and row-level security policies. This phase is documented in detail in Chapter 5 of this report.

e) Testing Phase

The testing phase, which occupies weeks ten and eleven of PSM 2, involves systematic verification that the implemented system meets the requirements and functions correctly. Unit testing is conducted on individual components including API endpoints, AI processing functions, and UI components. Integration testing is conducted on component interactions, such as the flow from mobile app upload through backend processing to database storage. User acceptance testing is conducted with sample users including tahfiz students and teachers to evaluate usability and satisfaction. Test cases are designed for each functional requirement, with expected results documented before testing begins. AI accuracy is evaluated against a test dataset of recitations with known correct pronunciation. Usability testing is conducted with observation and post-test questionnaires. This phase is documented in detail in Chapter 6 of this report.

f) Deployment Phase

The deployment phase, which occupies weeks twelve through fourteen of PSM 2, involves preparing the system for production use and documenting the final outcomes. Due to project timeline constraints, full-scale deployment to multiple tahfiz institutions is outside the scope of this project. However, a demonstration version is deployed to a cloud platform for evaluation by the supervisor and examiners. User documentation is prepared including installation instructions for the mobile app, user guides for both student and teacher interfaces, and technical documentation for future maintenance. The final report is completed and submitted, and the project is presented through an oral presentation and system demonstration.

2.4 Project Requirements

2.4.1 Software Requirement

The following software is required for the development and operation of the TasmiqAI system. Each software component was selected based on its suitability for the specific requirements of the project, as justified in the technique section (Section 2.2.3). Visual Studio Code serves as the primary integrated development environment for all coding activities, supporting Python, JavaScript, and TypeScript through extensions. Python version 3.9 or higher is required for the backend and AI engine, as this version provides the necessary language features and library compatibility for FastAPI, PyTorch, and Hugging Face Transformers. FastAPI version 0.100 or higher provides the RESTful API framework with automatic documentation and async support. Uvicorn serves as the ASGI server to run the FastAPI application in development and production environments. PyTorch version 2.0 or higher provides the deep learning framework for AI model inference, with support for both CPU and GPU execution. Hugging Face Transformers version 4.30 or higher provides access to pre-trained ASR models including Whisper and Wav2Vec2. Librosa version 0.10 or higher provides audio processing capabilities including silence trimming, noise reduction, and MFCC feature extraction. FFmpeg is required for audio codec conversion, particularly converting WebM format recordings from mobile devices to WAV format for processing. Node.js version 18 or higher provides the JavaScript runtime for frontend development, including both the React Native mobile application and the React teacher portal.

Expo CLI version 49 or higher provides the development and build toolchain for the React Native application, simplifying the process of running on physical devices. React Native version 0.72 or higher provides the mobile application framework. Vite version 4 or higher provides the build tool for the teacher portal, offering fast development server and optimized production builds. TailwindCSS version 3 or higher provides the utility-first CSS framework for styling the web portal. Git provides version control for source code management throughout the development process. GitHub serves as the remote repository for code backup and collaboration, and also serves as the source for AI model datasets and previous research code. Supabase provides the backend-as-a-service platform including PostgreSQL database, authentication, and storage. Anaconda provides Python environment management to isolate project dependencies and avoid conflicts with other Python projects on the development machine.

2.4.2 Hardware Requirement

The following hardware is required for development, testing, and deployment of the TasmiqAI system. For development hardware, a PC or laptop with Intel i7 or AMD Ryzen processor, 16 gigabytes of RAM, and 512 gigabyte solid-state drive is required to run Visual Studio Code and all development tools simultaneously, to execute the AI engine locally for testing, and to compile and build the mobile application. An optional NVIDIA GPU with 6 gigabytes or more of VRAM is recommended but not required, as it accelerates AI model inference by enabling GPU-based computation through PyTorch, significantly reducing the time required to process recitation audio files and generate feedback. For testing hardware, an Android smartphone running Android 10 or higher is required for mobile application testing, including testing the microphone for recitation recording and testing the user interface on a physical device. An iOS device such as an iPhone or iPad running iOS 14 or higher is also required for testing the iOS compatibility of the Expo application. A teacher PC or laptop with any modern processor and web browser is required to access the teacher portal for testing monitoring features. For student user hardware, a smartphone running Android 10 or higher or iOS 14 or higher is required to run the React Native and Expo mobile application. The built-in microphone on the smartphone is critical hardware for the system, as it captures the student's Quran recitation audio for AI assessment. Speaker or earphones, either built-in or external, are required to play reference recitations if implemented and to play back the student's own recitation for review. An internet connection via WiFi or cellular data is required to

upload recorded audio files to the backend server and to receive AI assessment results. For teacher user hardware, a PC, laptop, or tablet with any modern browser is required to access the React and Vite teacher portal. A display or monitor of any resolution is required to view dashboards, analytics charts, and student lists. Speakers or headphones, either built-in or external, are required to listen to student recitation audio files for manual review and verification. A keyboard and mouse are required to navigate the portal, input feedback scores, manage classes, and generate reports. For server hardware, a web server with 2 or more virtual CPUs and 4 gigabytes or more of RAM is required to host the FastAPI backend application. Note that Supabase handles database hosting as a cloud service, so no local database server hardware is required.

2.4.3 Other Requirements

The following additional requirements are necessary for the successful completion of this project. A stable internet connection is required throughout the development process for accessing GitHub repositories to download code and datasets, for downloading Hugging Face models which can be several gigabytes in size, for accessing Supabase cloud services during development and testing, and for accessing online documentation for all software tools and libraries. Access to Quran datasets is required, including Quranic audio datasets from Hugging Face and GitHub repositories for AI model testing and validation, as documented by Alam (2023) and Eldeeb (2023). Reference recitations are required, specifically high-quality Quran recitation audio files from professional qaris such as Sheikh Mishary Al-Afasy, to serve as comparison standards for evaluating AI assessment accuracy. Arabic text resources are required, specifically a digital copy of Quranic text with complete diacritics (tashkeel) to ensure accurate text matching between student recitation transcripts and reference verses. Access to university computer lab facilities is required for development and testing sessions, particularly for accessing GPU resources if available and for conducting user testing with tahfiz students. A discussion room is required for supervisor meetings and project presentations, providing a quiet space for progress reviews and feedback sessions. Printing facilities are required for report printing and documentation submission, including both draft versions for supervisor review and final copies for submission.

2.5 Project Schedule and Milestones

The project is divided into two semesters: Project Sarjana Muda 1 (PSM 1) which focuses on planning, analysis, design, and proposal documentation, and Project Sarjana Muda 2 (PSM 2) which focuses on implementation, testing, and final reporting. In PSM 1, during weeks one and two, project topic selection and supervisor meeting activities occurred, resulting in an approved topic. During weeks three and four, problem identification and objective definition activities occurred, resulting in documented problem statements and objectives. During weeks five and six, literature review and existing system analysis activities occurred, resulting in a completed literature review chapter. During weeks seven and eight, requirements analysis and methodology selection activities occurred, resulting in an analysis chapter. During weeks nine and ten, system design activities including architecture, database, and UI design occurred, resulting in a design chapter. During weeks eleven and twelve, proposal documentation and submission activities occurred, resulting in a complete proposal report. During weeks thirteen and fourteen, proposal presentation preparation activities occurred, resulting in presentation slides for the PSM 1 presentation. In PSM 2, during weeks one and two, development environment setup activities occurred, resulting in a configured development environment with all required software installed and verified. During weeks three through six, backend and AI engine implementation activities occurred, resulting in a functional FastAPI backend with authentication endpoints and a functional AI engine with speech recognition and Tajwid checking. During weeks seven through nine, mobile app and teacher portal implementation activities occurred, resulting in a functional React Native mobile application and a functional React teacher portal. During weeks ten and eleven, system testing and debugging activities occurred, resulting in documented test results and resolved bug fixes. During weeks twelve and thirteen, final report writing activities occurred, resulting in a complete final report. During week fourteen, final presentation and demonstration activities occurred, resulting in presentation slides and a working prototype for demonstration to examiners. A detailed Gantt chart illustrating the timeline of these activities is provided in Appendix B of this report.

2.6 Conclusion

This chapter has presented a comprehensive literature review of the domains relevant to the TasmiaqAI project, including Islamic education technology, automatic speech recognition for Arabic, and expert systems for rule-based error detection. The review of existing Quran learning applications, including Tarteel, Qara'a, TajweedMate, and the Tahfiz Intelligent System, revealed significant gaps in current solutions, including limited AI integration for detailed Tajwid checking, lack of teacher monitoring portals, proprietary technology that limits academic reproducibility, and scarcity of labeled Quranic datasets for research. The technique section evaluated alternative approaches for each system component, justifying the selection of Hugging Face Transformers over commercial APIs or custom models, FastAPI over Django or Flask, Supabase over Firebase or local MySQL, React Native with Expo over native or other cross-platform frameworks, and React with Vite over Create React App or Next.js. A limitation of the current system was transparently acknowledged: the AI speech recognition engine has limited accuracy for Quranic Arabic pronunciation, particularly for distinguishing similar phonemes, due to the inherent difficulty of the task, limited dataset availability, and computational constraints. The Waterfall SDLC methodology was described as the framework guiding this project through planning, analysis, design, implementation, testing, and deployment phases. Finally, the software, hardware, and other requirements were listed, followed by the project schedule and key milestones for both PSM 1 and PSM 2. The next chapter will present the analysis phase in detail, including problem analysis, requirement analysis covering data, functional, and non-functional requirements, and the justification for selected requirements.

CHAPTER 3. ANALYSIS

3.1 Introduction

This chapter presents a detailed analysis of the TasmiqAI system, translating the project requirements identified in Chapter 2 into concrete specifications that inform the design phase documented in Chapter 4. The analysis phase is a critical stage in the Waterfall software development methodology, as it bridges the gap between high-level project objectives and detailed system specifications. According to Pressman (2014), the analysis phase involves investigating the current system or situation, understanding user needs, and documenting requirements in a structured manner that can be verified and validated before design begins. This chapter is organized into three main sections. First, the problem analysis section investigates the current tahfiz education scenario, describing how tasmi' sessions are currently conducted and identifying specific pain points that TasmiqAI aims to address. Second, the requirement analysis section documents the data requirements, functional requirements, and non-functional requirements that the system must satisfy. Third, the conclusion summarizes the analysis findings and outlines the activities to be developed in the design phase.

3.2 Problem Analysis

3.2.1 Current System Scenario

To understand the problems that TasmiqAI aims to address, it is necessary to examine how Quran memorization and recitation assessment are currently conducted in typical tahfiz institutions. The traditional tasmi' (recitation testing) process follows a well-established pattern that has remained largely unchanged for centuries, relying on

direct oral transmission from teacher to student (Nasir & Abdul Rahman, 2020). Understanding this current scenario provides the basis for identifying opportunities where technology can enhance rather than replace the valuable aspects of traditional pedagogy. In a typical tahfiz class, students are organized into small groups under the supervision of a single teacher. Each student is expected to memorize a specific portion of the Quran, typically measured in terms of verses (ayat) or pages (muka surat). The memorization process involves repeated reading of the verses while listening to a teacher or recorded recitation, followed by self-practice, and finally recitation to the teacher for verification and correction (Mahmoud & Daud, 2021). This cycle repeats until the student demonstrates accurate memorization and correct pronunciation.

The tasmi' session itself follows a structured format. The student approaches the teacher, either individually or in a small group setting, and recites the memorized portion from memory without looking at the Quran text. The teacher listens attentively, noting any errors in pronunciation (tajwid), articulation points (makhraj), or verse sequence. When an error is detected, the teacher stops the student, provides correction, and may ask the student to repeat the verse correctly before continuing (Abdullah & Hassan, 2019). After the recitation is complete, the teacher provides an overall assessment, notes the student's progress in a record book, and assigns the next portion for memorization.

This traditional approach has several strengths that have ensured its continued use for centuries. The direct, personalized interaction between teacher and student allows for immediate, context-specific feedback that addresses the individual student's specific errors (Rahman et al., 2020). The teacher can observe not only auditory aspects of recitation but also physical indicators of proper makhraj, such as lip and tongue positioning. Furthermore, the teacher serves as a motivating presence, encouraging students who are struggling and maintaining accountability for consistent progress.

However, the traditional approach also exhibits significant limitations when implemented in modern educational contexts. The most pressing limitation is the student-to-teacher ratio, which in many tahfiz institutions ranges from 20:1 to 50:1 (Ismail & Zainal, 2021).

With such ratios, each student receives limited individual tasmi' time, typically only a few minutes per day or even per week. As noted by Nasir and Abdul Rahman (2020), students in large tahfiz classes may wait several days between individual tasmi' sessions, during which time they continue practicing independently without any feedback mechanism to catch errors that may be developing.

A second limitation concerns the documentation of student progress. Teachers typically maintain manual records using notebooks or spreadsheets, noting which verses each student has memorized, the scores achieved, and errors observed (Mahmoud & Daud, 2021). This documentation method is time-consuming, prone to errors or omissions, and does not facilitate systematic analysis of student performance over time. Teachers cannot easily identify patterns such as which Tajwid rules are most commonly violated by the class as a whole, or which students are consistently struggling with specific types of errors.

A third limitation relates to the cognitive load on teachers during tasmi' sessions. The teacher must simultaneously listen to the student's recitation, recall the correct pronunciation rules for each word being recited, compare the student's production to the correct standard, identify specific errors, decide whether to interrupt or wait until the verse is complete, formulate clear corrective feedback, and document the errors for future reference (Abdullah & Hassan, 2019). This cognitive multitasking is mentally demanding and can result in missed errors, inconsistent feedback, or teacher fatigue, particularly during long sessions with many students.

3.2.2 Detailed Problem Statement

Based on the analysis of the current tahfiz education scenario presented above, five specific problems have been identified that directly motivate the development of the TasmiqAI system. These problems are not merely theoretical concerns but represent practical challenges that affect tahfiz students and teachers on a daily basis.

Problem 1: Limited Access to Personalized Feedback During Self-Revision

Tahfiz students often practice recitation independently during muraja'ah (revision) sessions that occur outside of formal tasmi' hours. These self-directed practice sessions are essential for consolidating memorized portions and maintaining previously memorized Quran. However, during these independent practice sessions, students have no immediate way to verify the accuracy of their recitation, particularly regarding subtle Tajwid rules such as Idgham (merging of letters), Ikhfa (hiding of letters), and Mad (prolongation of vowel sounds). According to a study by Mahmoud and Daud (2021), approximately 65% of tahfiz students report practicing for at least one hour daily without any form of feedback on their recitation accuracy. Errors practiced repeatedly during these unsupervised sessions become ingrained through repetition, requiring significantly more effort to correct later when the error is finally identified during a formal tasmi' session with the teacher.

Problem 2: High Teacher Workload Limiting Individual Tasmi' Frequency

In typical tahfiz institutions, each teacher is responsible for supervising between 20 and 50 students. Conducting individual tasmi' sessions for every student on a daily basis is mathematically impractical, as each session would be limited to only a few minutes per student. Ismail and Zainal (2021) report that the average tahfiz student receives only 5 to 10 minutes of individual teacher attention per week for tasmi' assessment. As a result, each student receives limited direct teacher feedback, typically once or twice per week at most. This constraint significantly slows the learning progress, as students cannot receive timely correction on errors made during their daily independent practice. Furthermore, students who are struggling may fall behind without early intervention, as the infrequent tasmi' sessions may not detect problems until significant practice time has been invested in incorrect recitation patterns.

Problem 3: Lack of Systematic Progress Monitoring Tools

Teachers currently rely on manual records, such as notebooks or spreadsheets, to track student progress across multiple classes and over extended time periods. This approach

presents several significant disadvantages. First, manual documentation is time-consuming, requiring teachers to spend time writing or typing records that could otherwise be spent directly teaching students. Second, manual records are prone to errors, including transcription mistakes, lost pages, or illegible handwriting. Third, manual records do not facilitate data-driven analysis of student performance, as analyzing patterns across dozens of students and hundreds of recitation sessions would require manual compilation and calculation (Rahman et al., 2020). Teachers cannot easily identify class-wide error patterns, track individual student improvement over weeks or months, correlate performance with other variables, or generate performance reports for parents or administration without substantial manual effort.

Problem 4: Inefficient Error Identification During Tasmi' Sessions

Even when teachers conduct tasmi' sessions with individual students, the process of manually listening to each verse and identifying specific Tajwid errors is mentally demanding. Teachers must simultaneously perform multiple cognitive tasks: listening to the student's recitation, recalling the correct pronunciation rules for each word, comparing the student's production to the correct standard, identifying specific error types and locations, deciding on appropriate corrective feedback, and documenting the errors for future reference (Abdullah & Hassan, 2019). This cognitive load can result in several negative outcomes, including missed errors where the teacher does not notice a mistake, incomplete feedback where only the most obvious errors are addressed, and teacher fatigue that reduces the quality of instruction later in a session. Research by Nasir and Abdul Rahman (2020) suggests that even experienced tahfiz teachers may miss up to 15% of Tajwid errors during extended tasmi' sessions due to attention lapses or cognitive overload.

Problem 5: Limited Integration of Modern Technology in Quran Education

While many educational domains have enthusiastically embraced technology-assisted learning tools such as language learning applications, mathematics tutoring systems, and

automated writing feedback tools, Quran recitation learning remains largely traditional in its reliance on direct human instruction. A survey of existing Quran applications conducted by Salim et al. (2025) found that the majority of available applications focus either on displaying Quran text without pronunciation analysis or providing basic audio comparison features without proper Tajwid rule checking. This technology gap means that tahfiz students lack access to the kinds of automated feedback and progress tracking tools that students in other educational domains take for granted. Furthermore, the lack of technology integration means that tahfiz institutions cannot benefit from the data-driven insights and administrative efficiencies that educational technology can provide.

3.3 Requirement analysis

3.3.1 Data Requirement

The TasmiqAI system must store and manage various types of data to support its functionality. Understanding the data requirements is essential for designing an appropriate database schema that ensures data integrity, supports efficient queries, and scales to accommodate future growth. The following data entities have been identified as necessary for the system.

User Data: The system must store information about all registered users, including both students and teachers. For each user, the system must store a unique identifier, full name, email address for login and notifications, password hash for authentication, user role indicating whether the user is a student or teacher, registration date, and last login timestamp. For students, additional data includes class or group affiliation, overall progress percentage across all memorized surahs, and current learning goals. For teachers, additional data includes the classes or groups they supervise and their contact information.

Quran Reference Data: The system must store reference information about the Quran to enable surah and verse selection, text comparison, and progress tracking. This reference data includes surah names in Arabic and transliteration, surah numbers, verse numbers, verse text with full diacritics, and verse order within each surah. This reference data serves as the standard against which student recitations are compared.

Recitation Data: The system must store records of each recitation session initiated by a student. For each recitation, the system must store a unique session identifier, the student user identifier, the surah and verse range recited, the timestamp when the recitation was submitted, the audio file reference (path or URL to the stored audio file), the duration of the recitation in seconds, and the file size for storage management.

Assessment Data: For each recitation session, the system must store the AI-generated assessment results. This assessment data includes the overall accuracy score as a percentage, a structured representation of errors detected including error type, location within the verse, and severity, the transcribed text from the speech recognition engine, and the timestamp when the assessment was completed. For teacher-reviewed recitations, the teacher's score and comments are also stored.

Class and Group Data: The system must support the organization of students into classes or groups for teacher management. For each class, the system must store a class identifier, class name, teacher identifier for the supervising teacher, creation date, and current status. Student-class associations must be stored to indicate which students belong to which classes.

Feedback Data: The system must store feedback provided by teachers to students based on recitation assessments. For each feedback record, the system must store the feedback identifier, the student identifier, the teacher identifier, the recitation session identifier being commented on, the feedback text, and the timestamp when feedback was submitted.

System Logs: For security and debugging purposes, the system should store logs of

significant events including login attempts, recitation uploads, assessment completions, and error conditions. Each log entry includes a timestamp, event type, user identifier if applicable, and a detailed message.

3.3.2 Functional Requirement

Functional requirements specify what the system must do, describing the behaviors, operations, and functions that the system must support to meet user needs. According to Sommerville (2016), functional requirements should be specific, measurable, achievable, relevant, and testable. The following functional requirements have been identified for the TasmiqAI system, organized by subsystem.

User Authentication and Management Functions:

The system shall allow new users to register for an account by providing their name, email address, and password. The system shall validate that the email address is in a proper format and that the password meets minimum complexity requirements including minimum length and character type diversity. The system shall allow existing users to log in using their registered email address and password, and upon successful authentication, shall generate a JSON Web Token (JWT) for subsequent authenticated requests. The system shall restrict access to functions based on user role, such that students can only access student-specific functions and teachers can only access teacher-specific functions. The system shall allow users to reset their password if forgotten, by sending a password reset link to their registered email address.

Student Mobile Application Functions:

The system shall display a dashboard to students after login, showing their overall progress percentage across all surahs, their today's goal or recommended practice, and quick access buttons for starting a new recitation or viewing history. The system shall

allow students to select a surah and a verse range for recitation from a scrollable list of all 114 surahs. The system shall provide a recording interface that accesses the device microphone, allows the student to start and stop recording, displays the recording duration, and allows playback of the recording before submission. The system shall upload the recorded audio file to the backend server for processing, displaying a progress indicator during upload. The system shall retrieve and display AI assessment results including overall accuracy score, specific errors detected with locations, and suggested improvements. The system shall maintain a history of all previous recitations, allowing the student to view past scores, listen to past recordings, and see teacher feedback if provided.

Teacher Web Portal Functions:

The system shall display a dashboard to teachers after login, showing key metrics including total number of students across all classes, total number of recitations submitted, average class score, and number of recitations awaiting teacher review. The system shall allow teachers to view a list of all students they supervise, with basic information including student name, overall progress percentage, and last recitation date. The system shall provide a detailed student view showing all recitations submitted by that student, including dates, scores, AI assessments, and teacher feedback. The system shall allow teachers to play recitation audio files and review the AI assessment results, with the ability to override or modify the AI-generated score and add textual feedback comments. The system shall generate analytics reports showing performance trends over time, common error types across a class, and individual student progress compared to class averages.

Recitation Recording and Processing Functions:

The system shall accept audio file uploads from the mobile application in WebM or MP3 format through a REST API endpoint. The system shall validate uploaded files to ensure they are within size limits and are in an expected audio format before processing. The

system shall convert uploaded audio files to a standard format suitable for the AI engine, specifically 16kHz mono WAV. The system shall preprocess the audio to remove silence from beginning and end and apply noise reduction to improve speech recognition accuracy.

AI Assessment Functions:

The system shall pass preprocessed audio through an automatic speech recognition model to generate a text transcription of the student's recitation in Arabic. The system shall compare the transcribed text to the expected verse text from the Quran reference data, identifying missing verses, extra verses, or out-of-order recitation. The system shall apply a rule-based expert system to detect Tajwid errors including Makhraj (articulation point) errors, Mad (prolongation) errors where the duration of vowel sounds is incorrect, Idgham (merging) errors where letters that should be merged are pronounced separately, Ikhfa (hiding) errors, Izhar (clear pronunciation) errors, and Iqlab (conversion) errors. The system shall calculate an overall accuracy score as a percentage, weighting verse correctness and Tajwid rule compliance according to predefined weights. The system shall generate human-readable feedback describing detected errors and suggesting improvements.

Database Management Functions:

The system shall store all user data, recitation records, assessment results, class associations, and feedback in the Supabase PostgreSQL database. The system shall retrieve data efficiently using appropriate indexes on frequently queried fields such as user identifiers and timestamps. The system shall maintain referential integrity such that deleting a user also deletes associated recitation records or reassigns them appropriately.

3.3.3 Non-functional Requirement

Non-functional requirements specify how well the system performs its intended functions, addressing qualities such as performance, security, usability, and reliability (Sommerville, 2016). These requirements are essential for ensuring that the TasmiqAI system provides a satisfactory user experience and operates reliably in educational settings.

Performance Requirements:

The system shall process recitation audio files and return assessment results within 30 seconds for typical recitations of up to 60 seconds duration when running on a machine with GPU acceleration. For CPU-only processing, the system shall return results within 120 seconds. The mobile application shall respond to user interactions including button presses and screen navigation within 1 second. The teacher portal dashboard shall load and display data for a class of up to 50 students within 3 seconds. The system shall support concurrent usage by up to 100 students without significant degradation in response time.

Accuracy Requirements:

The AI speech recognition component shall achieve word error rate of no more than 25% on clear recitation audio from students with standard pronunciation. The Tajwid error detection component shall correctly identify at least 60% of actual Tajwid errors present in the recitation, with acknowledgement that perfect accuracy is not achievable with current models as discussed in Section 2.2.3. The system shall provide confidence scores with each assessment, indicating the AI's certainty in its analysis, to help teachers and students appropriately weigh the automated feedback.

Security Requirements:

The system shall store user passwords using strong cryptographic hashing with salt to

prevent password recovery even if the database is compromised. All communication between the mobile application, teacher portal, and backend server shall be encrypted using HTTPS to prevent eavesdropping on network traffic. User authentication shall use JSON Web Tokens (JWT) with appropriate expiration times to limit the window for token theft. The system shall implement row-level security policies in Supabase such that teachers can only access data for students in their own classes, not for students supervised by other teachers. Audio files containing student recitations shall be stored with access controls limiting retrieval to the student who recorded them and teachers who supervise that student.

Usability Requirements:

The mobile application user interface shall be usable by tahfiz students aged 10 years and older without requiring technical training beyond basic smartphone operation. All user interface text shall be available in both English and Malay languages, with the ability to switch between languages without restarting the application. The recording interface shall provide clear visual feedback indicating that recording is in progress, the duration recorded so far, and confirmation when recording is complete. Error messages shall be written in clear, non-technical language and shall suggest how to resolve the error when possible.

Reliability and Availability Requirements:

The system shall maintain uptime of at least 99% during normal operating hours (8 AM to 8 PM) when deployed to cloud infrastructure. The system shall handle network interruptions gracefully, allowing students to save recordings locally for later upload when connectivity is restored. The system shall log errors for debugging and shall not crash or lose data when unexpected inputs are received.

Scalability Requirements:

The system architecture shall support scaling to accommodate additional surahs and

verses beyond the initial implementation, which already covers all 114 surahs. The database schema shall support expansion to additional features such as parent accounts, school administration dashboards, or advanced analytics without requiring complete redesign.

3.3.4 Others Requirement

Beyond the functional and non-functional requirements specified above, several additional requirements have been identified that pertain to the development process and deployment context rather than the system's runtime behavior.

Development Environment Requirements:

The development environment shall use Visual Studio Code as the primary integrated development environment. All source code shall be managed using Git for version control, with a GitHub repository serving as the remote backup and collaboration platform. The backend and AI engine shall be developed using Python 3.9 or higher with dependencies managed through pip and Conda. The mobile application shall be developed using React Native with Expo, running on either physical Android or iOS devices for testing. The teacher portal shall be developed using React with Vite and TailwindCSS, running on the development machine with hot reload enabled.

Deployment Requirements:

The system shall be deployable to cloud platforms including Render, Vercel, or similar services for demonstration purposes. Environment variables shall be used for configuration including API keys and database connection strings, never hardcoded in source code. The database schema shall be version controlled using Supabase migrations to enable consistent updates across development and deployment environments.

Documentation Requirements:

User documentation shall be provided including a student guide explaining how to register, select surahs, record recitations, and understand feedback, and a teacher guide explaining how to monitor students, review recitations, and provide feedback. Technical

documentation shall be provided including API endpoint documentation generated automatically by FastAPI, database schema documentation, and instructions for setting up the development environment. The final project report shall follow the formatting guidelines specified in the Final Year Project Writing Guide 2025.

3.4 Conclusion

This chapter has presented a detailed analysis of the TasmiqAI system, building on the problem statements and objectives defined in Chapter 1. The problem analysis section investigated the current tahfiz education scenario, describing how traditional tasmi' sessions are conducted and identifying five specific problems: limited access to feedback during self-revision, high teacher workload limiting individual tasmi' frequency, lack of systematic progress monitoring tools, inefficient error identification during tasmi' sessions, and limited integration of modern technology in Quran education. The requirement analysis section documented the data requirements for user data, Quran reference data, recitation data, assessment data, class data, feedback data, and system logs. Functional requirements were specified for user authentication, student mobile application functions, teacher web portal functions, recitation recording and processing, AI assessment, and database management. Non-functional requirements were specified for performance, accuracy, security, usability, reliability, and scalability. Other requirements including development environment, deployment, and documentation were also identified.

The analysis presented in this chapter provides the foundation for the design phase documented in Chapter 4. The next chapter will translate these requirements into concrete system designs including high-level architecture, database schema, user interface designs, and detailed component specifications.

CHAPTER 4. DESIGN

4.1 Introduction

This chapter presents the design phase of the TasmiqAI system, translating the requirements documented in Chapter 3 into concrete system specifications that guide the implementation phase documented in Chapter 5. According to Pressman (2014), the design phase is the point at which the "what" of the requirements is transformed into the "how" of the system architecture, user interfaces, database structures, and detailed component specifications. This chapter is organized into seven main sections. First, the high-level design section provides an overview of the system's structural organization. Second, the system architecture section defines the architectural view of the system using appropriate diagrams. Third, the user interface design section presents navigation design, input design, and output design for both the mobile application and teacher web portal. Fourth, the database design section presents the conceptual, logical, and physical database designs including entity-relationship diagrams and data dictionaries. Fifth, the detailed design section elaborates on the software design for each component. Sixth, the physical database design section translates the logical design to the target DBMS. Seventh, a conclusion summarizes the design activities and outlines the next phases.

4.2 High-Level Design

The high-level design of the TasmiqAI system provides a bird's-eye view of the system's structure and the organization of its major components. According to Sommerville

(2016), high-level design should identify the major subsystems or modules of the system and define the relationships and interactions between them.

The TasmiqAI system is organized into four major subsystems, each responsible for a distinct set of functions:

1. **Student Mobile Application Subsystem:** This subsystem encompasses all user interface components and client-side logic for the student-facing mobile application. It is responsible for user authentication, surah and verse selection, audio recording, upload management, feedback display, and progress tracking. The mobile application communicates with the backend subsystem through REST API calls.
2. **Teacher Web Portal Subsystem:** This subsystem encompasses all user interface components for the teacher-facing web portal. It is responsible for teacher authentication, dashboard display, student management, recitation review, feedback submission, analytics generation, and class management. The web portal communicates with the backend subsystem through REST API calls.
3. **Backend and AI Processing Subsystem:** This subsystem is the core of the TasmiqAI system, handling all business logic and AI processing. It is responsible for user authentication and authorization, recitation upload handling, audio preprocessing, speech recognition using Hugging Face Whisper, Tajwid rule checking using the expert system, score calculation, feedback generation, and database interaction. This subsystem is implemented as a FastAPI application with integrated AI modules.
4. **Database Subsystem:** This subsystem manages all persistent data storage for the system. It is responsible for storing user profiles, recitation records, assessment results, class associations, and teacher feedback. The database subsystem is implemented using Supabase, which provides a PostgreSQL database with built-in authentication and row-level security.

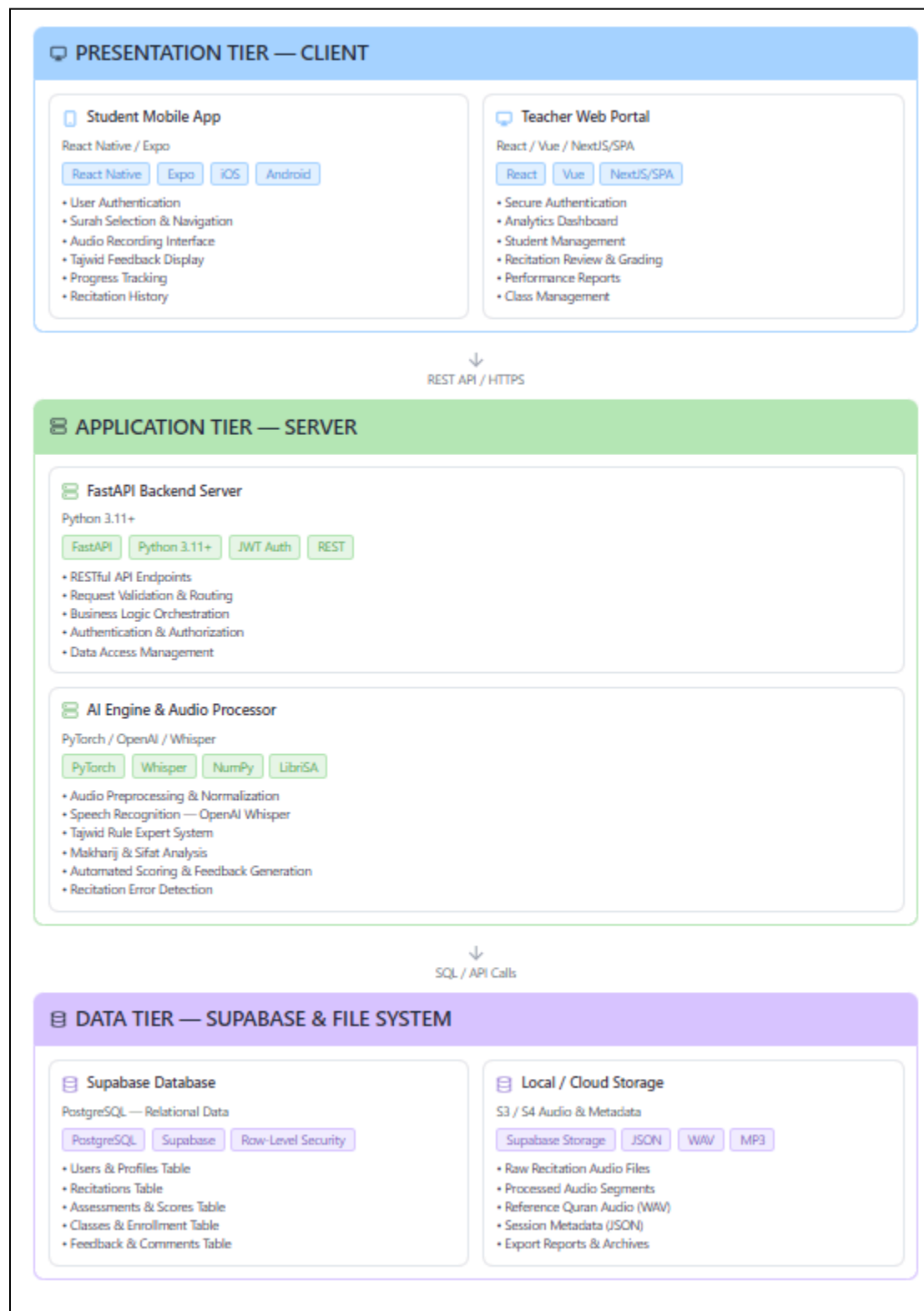


Figure 4.1 illustrates the high-level architecture of the TasmiqAI system. The diagram

shows the three-layer organization: the presentation layer containing the student mobile application and teacher web portal, the application layer containing the FastAPI backend and AI engine, and the data layer containing the Supabase PostgreSQL database. Arrows indicate the direction of communication between layers.

4.3 System Architecture

The system architecture of TasmiqAI defines the detailed structure of the system, including the static view of components and the dynamic view of interactions between components. The architecture follows a client-server model with a decoupled three-tier design, ensuring that each layer can be developed, tested, and deployed independently.

Figure 4.2: Detailed System Architecture with Data Flow

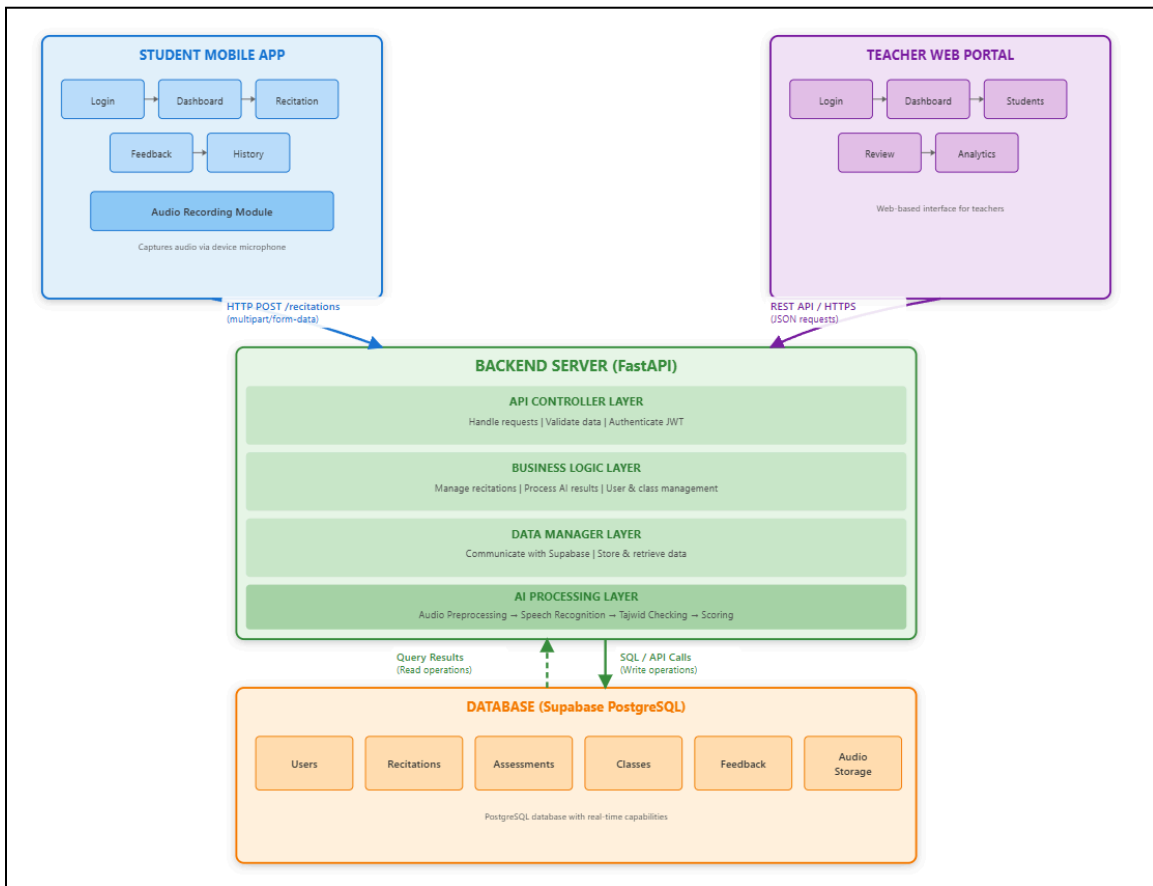


Figure 4.2 presents the detailed system architecture of TasmiqAI. The diagram shows all major components and the data flow between them. The student mobile application captures audio and sends it to the backend server via HTTP POST requests. The backend server processes the audio through the AI engine pipeline, which includes audio preprocessing, speech recognition, Tajwid checking, and scoring. Results are stored in the Supabase database and returned to the mobile application. The teacher web portal retrieves data from the database to display dashboards, student details, and recitation reviews.

Static View Components: The static view of the system consists of the following major components with their responsibilities:

- **MobileApp Component:** Handles user interface rendering, user input capture, audio recording, API communication, and local data caching.
- **WebPortal Component:** Handles teacher interface rendering, dashboard data visualization, audio playback, and feedback submission.
- **BackendAPI Component:** Exposes REST endpoints, validates requests, authenticates users, orchestrates business logic, and manages database transactions.
- **AIEngine Component:** Encapsulates the audio preprocessing, speech recognition model loading, Tajwid rule evaluation, and score calculation logic.
- **Database Component:** Stores persistent data and provides query capabilities through the Supabase client.

Dynamic View Interactions: The dynamic view of the system describes the sequence of interactions during key use cases. For the recitation submission use case, the sequence is as follows: (1) Student selects surah and verse range in the mobile app, (2) Student records audio using the device microphone, (3) Mobile app sends HTTP POST request to `/api/recitations` endpoint with audio file attached, (4) Backend validates the request and authenticates the user, (5) Backend saves audio file to temporary storage, (6) Backend calls AI engine to process the audio, (7) AI engine preprocesses audio, runs Whisper transcription, applies Tajwid rules, calculates score, (8) Backend saves recitation record and assessment results to database, (9) Backend returns assessment results to mobile app as JSON, (10) Mobile app displays feedback to student.

4.4 User Interface Design

The user interface design for TasmiqAI encompasses both the student mobile application and the teacher web portal. According to Shneiderman and Plaisant (2010), effective user interface design should follow principles of consistency, feedback, and user control, ensuring that users can accomplish their tasks efficiently and with minimal errors.

4.4.1 Navigation Design

The navigation design defines how users move between screens or pages within the application. Both the mobile application and teacher portal are designed with intuitive navigation structures that reflect the primary tasks of each user role.

Figure 4.3: Mobile Application Navigation Flow

diagram here

Figure 4.3 illustrates the navigation flow for the student mobile application. The flow begins at the splash screen, branching to either login or registration for unauthenticated users. After authentication, the dashboard serves as the central hub, providing access to the recitation screen, learning mode selection, and history screen. The recitation flow proceeds through surah selection, recording, and feedback display, with the option to return to the dashboard or record another recitation.

Figure 4.4: Teacher Portal Navigation Flow

diagram here

Figure 4.4 illustrates the navigation flow for the teacher web portal. The flow begins at the login screen, leading to the dashboard as the central hub. From the dashboard, teachers can navigate to the students list, pending recitations, or analytics reports. The student detail view provides comprehensive information about an individual student, while the recitation review screen allows teachers to listen to recordings and provide feedback.

4.4.2 Input Design

The input design specifies the types of inputs required from users and the validation rules applied to each input field. Proper input design reduces errors and improves user experience by preventing invalid data from being submitted.

Student Mobile Application Inputs:

Input Field	Input Type	Validation Rules
Email (Login)	Text (keyboard)	Must be valid email format (e.g., user@example.com); required

Password (Login)	Text (password)	Minimum 6 characters; required
Name (Registration)	Text	Letters, spaces only; 2-50 characters; required
Email (Registration)	Text	Valid email format; must be unique; required
Password (Registration)	Text	Minimum 6 characters; must match confirmation; required
Surah Selection	Dropdown/Scroll	Must select from available surahs (1-114); required
Verse Range	Number pickers	Start verse $\leq$ End verse; within surah limits; required
Audio Recording	Microphone input	Minimum 1 second; maximum 120 seconds; required before submit

Teacher Web Portal Inputs:

Input Field	Input Type	Validation Rules
Email	Text	Valid email format; required
Password	Text	Minimum 6 characters; required
Student Search	Text	Optional; free text search
Feedback Text	Textarea	1-1000 characters; optional
Score Override	Number	0-100; optional; only if overriding AI score

Class Name	Text	1-50 characters; required for class creation
Student Assignment	Checkboxes	Select from existing students; at least one required

4.4.3 Output Design

The output design specifies the types of information presented to users, including dashboards, feedback displays, and reports. Outputs are designed to be clear, informative, and actionable.

Student Mobile Application Outputs:

Output Type	Format	Frequency	Description
Dashboard Progress	Percentage bar + text	Real-time (on load)	Shows overall memorization progress across all surahs
Today's Goal	Text + verse reference	Daily (on dashboard)	Suggests which verses to practice based on history
Recitation Score	Percentage + color code	After each submission	0-100% with green (80-100), yellow (60-79), red (0-59)
Error Feedback	Highlighted text + descriptions	After each submission	Specific errors with location and correction suggestion
Recitation History	List with scores and dates	On History screen	Chronological list of past recitations with playback

Learning Mode Selection	Card-based options	On Learning screen	Murajah Mode, New Memorization, Revision options
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Teacher Web Portal Outputs:

Output Type	Format	Frequency	Description
Dashboard Summary	Cards with numbers	Real-time (on load)	Total students, average score, pending reviews
Student List	Table with pagination	Real-time (on load)	Names, progress percentages, last activity
Student Detail	Profile view with tabs	On student selection	Full history, performance trends, feedback log
Recitation Review	Audio player + score + feedback	On recitation selection	Play audio, view AI assessment, add teacher feedback
Class Analytics	Bar charts + line charts	On demand	Performance distribution, error type frequency
Reports	PDF download	On demand	Individual or class performance reports

4.5 Database Design

The database design for TasmiqAI follows a structured approach beginning with conceptual modeling, proceeding to logical design, and concluding with physical implementation. According to Elmasri and Navathe (2016), a well-designed database ensures data integrity, supports efficient querying, and scales to accommodate growth.

4.5.1 Conceptual and Logical Database Design

The conceptual database design identifies the main entities in the TasmiqAI system and their relationships, independent of any specific database management system. The logical database design translates this conceptual model into a relational schema with tables, columns, data types, and constraints.

Entities Identified:

1. User - Represents both students and teachers, with a role attribute distinguishing between them
2. Recitation - Represents a single recitation session submitted by a student
3. Assessment - Represents the AI-generated assessment results for a recitation
4. Class - Represents a group of students under a teacher's supervision
5. ClassMember - Represents the many-to-many relationship between students and classes
6. Feedback - Represents teacher feedback provided on a recitation

Business Rules:

- A User can be either a student or a teacher, not both (mutually exclusive roles)
- A Teacher can supervise multiple Classes
- A Class can have multiple Students (through ClassMember)
- A Student can belong to multiple Classes (through ClassMember)
- A Student can submit multiple Recitations
- Each Recitation has exactly one Assessment
- A Teacher can provide multiple Feedback entries
- Each Feedback is associated with exactly one Recitation and one Teacher

4.5.2 Entity Relationship Diagram (ERD)

Figure 4.5: Entity Relationship Diagram (ERD) for TasmiqAI Database

diagram here

Figure 4.5 presents the Entity Relationship Diagram for the TasmiqAI database. The diagram shows six main entities: User, Recitation, Assessment, Class, ClassMember, and Feedback. The User entity has a one-to-many relationship with Recitation (one user can have many recitations). Each Recitation has exactly one Assessment. The Class entity has a many-to-many relationship with User (student) through the ClassMember junction table, and a one-to-many relationship with User (teacher). Feedback connects Recitation with Teacher users.

4.5.3 Data Dictionary

The data dictionary provides detailed descriptions of each table, column, data type, and constraint in the database schema.

Table 4.1: Users Table

Column	Data Type	Constraints	Description
id	UUID	PRIMARY KEY, DEFAULT gen_random_uuid()	Unique identifier for each user
email	VARCHAR(255)	NOT NULL, UNIQUE	User email address used for login
password_hash	VARCHAR(255)	NOT NULL	Hashed password using bcrypt algorithm
full_name	VARCHAR(100)	NOT NULL	User's full name
role	user_role_enum	NOT NULL	'student' or 'teacher'
progress_percentage	DECIMAL(5,2)	DEFAULT 0, student only	Overall memorization progress (0-100)
created_at	TIMESTAMP	DEFAULT NOW()	Account creation timestamp
last_login	TIMESTAMP	NULL	Last login timestamp
class_id	UUID	FOREIGN KEY REFERENCES classes(id)	Class membership (alternative to junction table)

Table 4.2: Recitations Table

Column	Data Type	Constraints	Description
id	UUID	PRIMARY KEY, DEFAULT gen_random_uuid()	Unique recitation session identifier
user_id	UUID	FOREIGN KEY REFERENCES users(id), NOT NULL	Student who submitted the recitation
surah_number	INTEGER	NOT NULL, CHECK (1-114)	Surah number of recited verses
start_verse	INTEGER	NOT NULL, CHECK (≥ 1)	Starting verse number
end_verse	INTEGER	NOT NULL, CHECK ($\geq$ start_verse)	Ending verse number
audio_url	VARCHAR(500)	NOT NULL	Supabase storage path to audio file
submitted_at	TIMESTAMP	DEFAULT NOW()	Submission timestamp
duration_seconds	INTEGER	NOT NULL, CHECK (> 0)	Length of recitation in seconds
device_info	JSONB	NULL	Optional device metadata

Table 4.3: Assessments Table

Column	Data Type	Constraints	Description
id	UUID	PRIMARY KEY, DEFAULT gen_random_uuid()	Unique assessment identifier

recitation_id	UUID	FOREIGN KEY REFERENCES recitations(id), UNIQUE, NOT NULL	Associated recitation
score	INTEGER	NOT NULL, CHECK (0-100)	Overall accuracy score (0-100)
transcript	TEXT	NULL	Whisper-generated Arabic transcription
errors_json	JSONB	NOT NULL	Structured error data with types and locations
feedback_text	TEXT	NULL	Human-readable feedback summary
confidence_score	DECIMAL(3,2)	NULL, CHECK (0-1)	AI confidence in assessment (0-1)
assessed_at	TIMESTAMP	DEFAULT NOW()	Assessment completion timestamp

Table 4.4: Classes Table

Column	Data Type	Constraints	Description
id	UUID	PRIMARY KEY, DEFAULT gen_random_uuid()	Unique class identifier
name	VARCHAR(100)	NOT NULL	Class name (e.g., "Tahfiz 5A")
teacher_id	UUID	FOREIGN KEY REFERENCES users(id), NOT NULL	Teacher supervising the class

created_at	TIMESTAMP	DEFAULT NOW()	Class creation timestamp
description	TEXT	NULL	Optional class description

Table 4.5: Feedback Table

Column	Data Type	Constraints	Description
id	UUID	PRIMARY KEY, DEFAULT gen_random_uuid()	Unique feedback identifier
recitation_id	UUID	FOREIGN KEY REFERENCES recitations(id), UNIQUE, NOT NULL	Recitation being reviewed
teacher_id	UUID	FOREIGN KEY REFERENCES users(id), NOT NULL	Teacher providing feedback
feedback_text	TEXT	NOT NULL	Teacher's written feedback
override_score	INTEGER	NULL, CHECK (0-100)	Teacher's score overriding AI score
created_at	TIMESTAMP	DEFAULT NOW()	Feedback submission timestamp

4.5.4 Normalization

The database schema was normalized to third normal form (3NF) to eliminate redundancy and ensure data integrity. According to Elmasri and Navathe (2016), normalization is the process of organizing data to minimize redundancy and dependency by decomposing tables appropriately.

First Normal Form (1NF): All tables have atomic values in each column. For example,

the `errors_json` column in the `Assessments` table stores structured JSON rather than comma-separated values, and each recitation has its own row rather than storing multiple recitations in a single row.

Second Normal Form (2NF): All non-key attributes are fully functionally dependent on the primary key. In the `Recitations` table, columns such as `surah_number`, `start_verse`, and `end_verse` depend on the entire recitation ID, not just part of it.

Third Normal Form (3NF): No transitive dependencies exist where non-key attributes depend on other non-key attributes. For example, the teacher's name is not stored in the `Classes` table; instead, only the `teacher_id` foreign key is stored, and the teacher's name is retrieved through a join with the `Users` table when needed.

4.6 Detailed Design

The detailed design elaborates on the logic of each system component, providing specifications that guide implementation. According to Pressman (2014), detailed design should describe the internal logic of each function, including algorithms, data structures, and interfaces.

4.6.1 Software Design (Backend - FastAPI)

The FastAPI backend is organized into three layers: API controllers (routes), business logic services, and data access managers. Each layer has specific responsibilities and communicates with adjacent layers through well-defined interfaces.

Figure 4.6: FastAPI Backend Module Structure

diagram here

Figure 4.6 illustrates the module structure of the FastAPI backend. The application entry point (`main.py`) initializes the FastAPI app and registers routers. The API routers layer handles HTTP requests and responses. The business logic services layer contains the core application logic. The data managers layer abstracts database operations.

Table 4.6: API Endpoints Design

Endpoint	Method	Request Body	Response	Description
/api/auth/register	POST	{email, password, full_name, role}	{user_id, message}	Register new user
/api/auth/login	POST	{email, password}	{access_token, token_type}	Authenticate user
/api/recitations	POST	multipart/form-data: audio_file, surah_number, start_verse, end_verse	{recitation_id, assessment}	Submit recitation
/api/recitations/user/{user_id}	GET	-	{recitations: []}	Get user's recitations
/api/assessments/{recitation_id}	GET	-	{score, errors, feedback}	Get assessment
/api/teachers/dashboard	GET	-	{stats, recent_recitations}	Teacher dashboard data
/api/teachers/students	GET	-	{students: []}	Get teacher's students
/api/teachers/feedback	POST	{recitation_id, feedback_text, override_score}	{feedback_id}	Submit teacher feedback
/api/classes	GET	-	{classes: []}	Get teacher's classes
/api/classes	POST	{name, description}	{class_id}	Create new class

4.6.2 Software Design (AI Engine)

The AI engine is the core processing component of TasmiqAI, responsible for converting

raw audio into structured assessment results. The engine follows a pipeline architecture where audio passes through sequential processing stages.

Figure 4.7: AI Engine Processing Pipeline

diagram here

Figure 4.7 illustrates the AI engine processing pipeline. The pipeline consists of five sequential stages: audio preprocessing, speech recognition using Whisper, text comparison with expected verses, Tajwid rule checking using the expert system, and scoring with feedback generation.

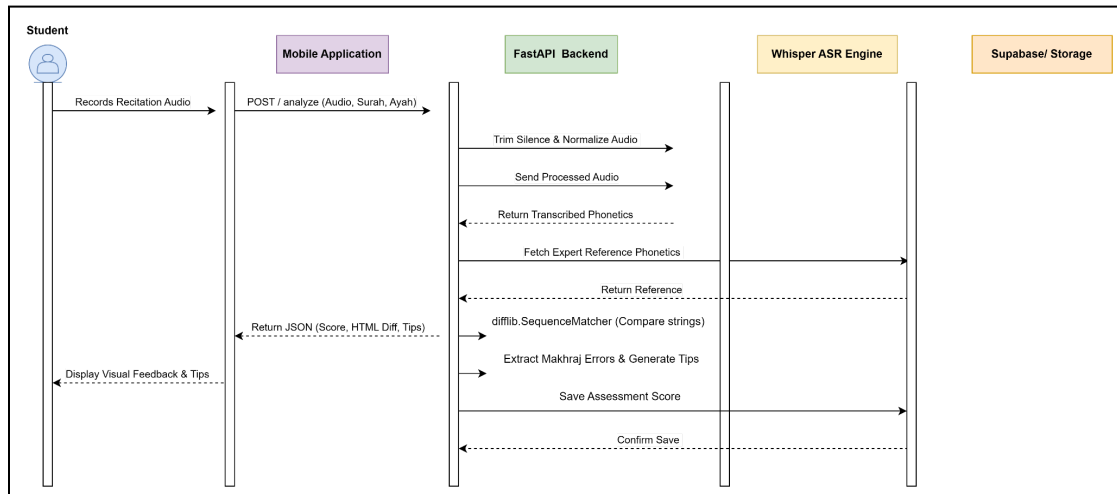
4.6.3 Software Design (Mobile Application)

The React Native mobile application follows a component-based architecture using functional components and React hooks. The main screens and their responsibilities are described below.

Component	File	Description
App.tsx	Root	Navigation container, authentication state provider
LoginScreen	screens/LoginScreen.tsx	Email/password login form, API call to /auth/login
RegisterScreen	screens/RegisterScreen.tsx	Registration form with validation
DashboardScreen	screens/DashboardScreen.tsx	Displays progress, today's goal, navigation menu
RecitationScreen	screens/RecitationScreen.tsx	Surah selection, verse range picker, recording controls
FeedbackScreen	screens/FeedbackScreen.tsx	Displays score, error list, improvement suggestions
HistoryScreen	screens/HistoryScreen.tsx	List of past recitations with scores and playback

Key Hooks and Services:

Code



4.6.4 Software Design (Teacher Portal)

The React teacher portal follows a similar component-based architecture with additional complexity for data visualization and table management.

Component	File	Description
App.tsx	Root	Router configuration, protected routes
Dashboard	pages/Dashboard.tsx	Statistics cards, recent recitations chart
StudentsList	pages/StudentsList.tsx	Paginated table of students with search
StudentDetail	pages/StudentDetail.tsx	Student profile, recitation history, progress chart
RecitationReview	pages/RecitationReview.tsx	Audio player, AI score display, feedback form

Analytics	pages/Analytics.tsx	Performance charts, error distribution
ClassManagement	pages/ClassManagement.tsx	Class CRUD operations, student assignment

4.7 Physical Database Design

The physical database design translates the logical schema to the target DBMS (Supabase PostgreSQL) with specific data types, indexes, and storage configurations.

Table Creation SQL (Supabase):

code

File Organization and Indexes:

Table	Indexes	Purpose
users	idx_users_email, idx_users_role	Fast login by email; filter by role
recitations	idx_recitations_user_id, idx_recitations_submitted_at	User history queries; time-based sorting
assessments	idx_assessments_recitation_id	Fast lookup by recitation
classes	idx_classes_teacher_id	Teacher's class list
class_members	idx_class_members_class_id, idx_class_members_student_id	Join performance

Storage Configuration (Supabase Storage):

- Bucket name: recitation-audio
- File path pattern:

`{user_id}/{surah_number}_{start_verse}_{end_verse}_{timestamp}.webm`

- File size limit: 10MB per file
- Access policy: Authenticated users can upload their own files; teachers can download files for students in their classes

4.8 Conclusion

This chapter has presented the design phase of the TasmiqAI system, translating the requirements from Chapter 3 into concrete design specifications for implementation. The high-level design section provided a bird's-eye view of the system's organization into four major subsystems: student mobile application, teacher web portal, backend and AI processing, and database. The system architecture section defined the static view of components and the dynamic view of interactions, illustrated through detailed architecture diagrams.

The user interface design section presented navigation flows for both the mobile application and teacher portal, input validation rules, and output specifications including dashboards, feedback displays, and reports. The database design section presented the entity-relationship diagram showing six main entities and their relationships, data dictionaries for each table, and normalization to third normal form.

The detailed design section elaborated on the FastAPI backend module structure, API endpoints, the AI engine processing pipeline with pseudocode, mobile application component specifications, and teacher portal component specifications. The physical database design section provided SQL table creation statements, indexing strategies, and storage configurations for Supabase.

The next chapter will present the implementation phase, documenting the setup of the development environment, software configuration management, version control procedures, and the implementation status of each module.

CHAPTER 5. IMPLEMENTATION

5.1 Introduction

This chapter details the transition from the system design phase into the physical development of the TasmiqAI system. The implementation phase represents the point at which the theoretical designs and architectural diagrams documented in Chapter 4 are converted into executable code, deployable software components, and a functioning database schema. According to Pressman (2014), successful implementation requires careful attention to environment configuration, version control practices, and systematic documentation of development progress. This chapter covers the setup of the development environments, the hardware and software architecture deployed for development and testing, configuration management strategies, version control procedures, and the current implementation status of each core module within the TasmiqAI system.

5.2 Software Development Environment setup

The development environment for TasmiqAI was structured around a decoupled, three-tier architecture to ensure scalability, maintainability, and ease of deployment across different computing platforms. As noted by Sommerville (2016), a well-designed development environment reduces integration problems and allows developers to focus on implementing features rather than resolving environment-specific issues.

5.2.1 Deployment and Environment Architecture

The architecture of the TasmiqAI development environment consists of three primary tiers: the client tier, the application tier, and the data tier. The client tier encompasses the student mobile application and the teacher web portal, both of which interact with

backend services through RESTful API calls. The application tier comprises the FastAPI backend server and the AI engine with audio processing capabilities. The data tier includes the Supabase PostgreSQL database and the storage system for audio reference files.

Figure 5.1: Deployment Diagram Showing Physical Hardware, Software Servers, and Network Connections

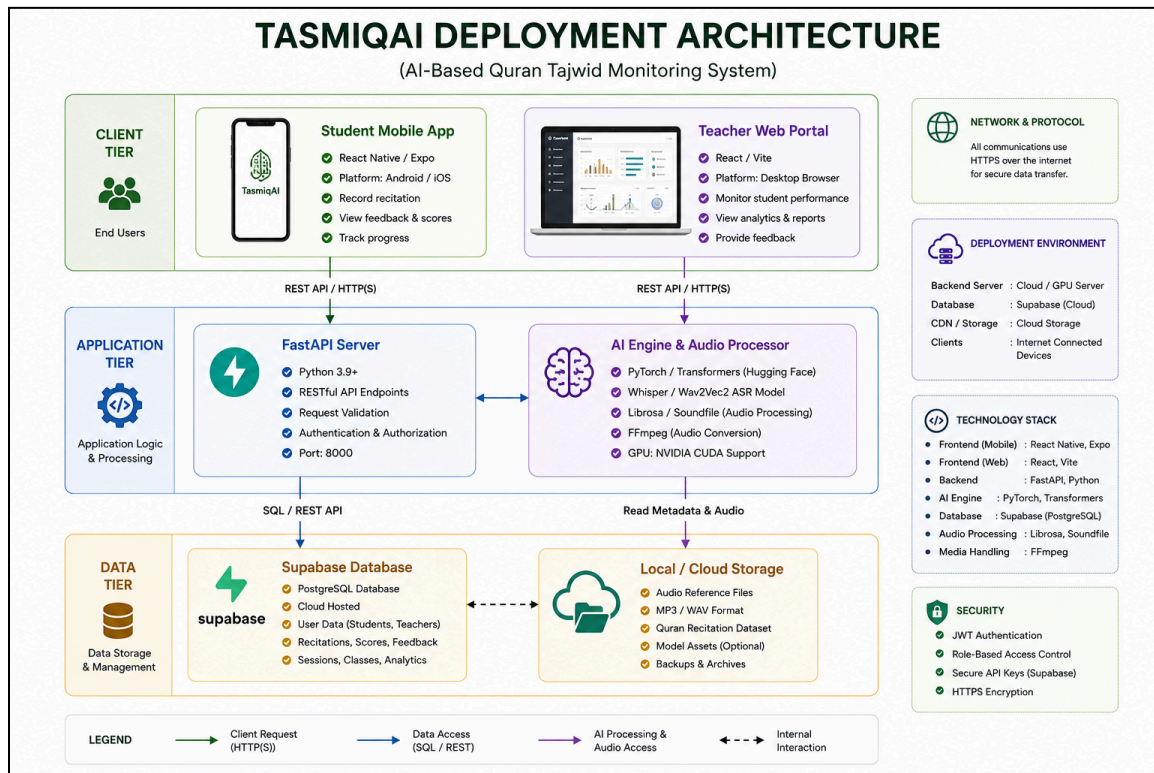


Figure 5.1 illustrates the deployment architecture of the TasmiqAI system. The diagram presents the three-tier architecture comprising the client tier (mobile and web applications), the application tier (FastAPI backend and AI engine), and the data tier (Supabase database and audio storage). The arrows indicate the direction of data flow and communication protocols between tiers.

5.2.2 Hardware and Network Setup

The hardware and network infrastructure for the TasmiqAI system was configured to support efficient development, testing, and eventual deployment. Each component was

selected based on the computational requirements of the AI processing tasks and the need for responsive user interactions.

The backend inference server was implemented on a dedicated machine equipped with an NVIDIA GPU supporting CUDA acceleration. This hardware configuration is essential for ensuring low-latency processing of the Hugging Face Whisper automatic speech recognition model, as deep learning inference on CPU-only systems would result in unacceptable response times for student users. As documented by Radford et al. (2023), the Whisper model benefits significantly from GPU acceleration, with inference times reduced by approximately 80% compared to CPU execution. The server operates on port 8000 to serve API requests from both the mobile application and teacher portal.

The database tier is hosted externally as a Supabase cloud service, accessed securely via API keys. This cloud-hosted approach eliminates the need for local database server maintenance while providing reliable, scalable storage for relational data including user profiles, recitation scores, and session history. Access to the database is restricted through API keys and row-level security policies, ensuring that only authenticated requests from the backend server can modify or retrieve data.

Client devices accessing the TasmiqAI system include student smartphones running either Android or iOS operating systems, and teacher desktop or laptop computers equipped with standard web browsers. Both client types require an active internet connection to communicate with the backend server, as all AI processing is performed server-side rather than on the client device.

5.2.3 Software Configuration

The software configuration for the TasmiqAI system was established with careful attention to dependency management, environment isolation, and developer productivity. Each component was configured following best practices documented in the respective framework documentation.

The backend was configured within Visual Studio Code as the primary integrated

development environment. A Python virtual environment (`venv`) was created to isolate project dependencies and avoid conflicts with other Python projects on the development machine. Key dependencies installed within this virtual environment include `fastapi` for the web framework, `torch` for deep learning operations, `transformers` for access to Hugging Face pre-trained models, and `librosa` for audio processing. Additionally, FFmpeg was explicitly installed in the system PATH to enable audio codec conversions, particularly converting WebM format recordings from mobile devices to WAV format suitable for the AI engine.

The frontend configuration was built using Node.js as the JavaScript runtime environment. The mobile application utilizes the Expo CLI, which provides a streamlined development experience by enabling rapid testing on physical devices through QR code scanning. Expo eliminates the need for native compilation during development, significantly accelerating the feedback loop for UI changes. The teacher web portal uses Vite as the build tool, which provides optimized Hot Module Replacement (HMR) during development, allowing changes to be reflected in the browser instantly without full page reloads.

5.3 Software Configuration Management

5.3.1 Configuration Environment Setup

The TasmicAI project follows a monorepo-style logical structure, where all code for different system components resides within a single repository but is organized into distinct modular directories. This approach, as described by Chacon and Straub (2014), allows independent development and testing of components while maintaining a unified version history. The directory structure is organized as follows:

- `tasmic_api.py` and `tasmic_app.py` contain the core AI logic and FastAPI backend implementation, serving as the entry points for the application tier.
- `tasmic-mobile/` contains the student Expo application, including screens,

- components, and navigation logic.
- `tasmiq-teacher-portal/` contains the React Vite web portal for teacher monitoring and administration.
 - `deps/` contains external binaries and dependencies such as FFmpeg that are required for system operation but are not Python packages.

This modular organization enables parallel development of different system components without creating merge conflicts, as changes to the mobile application do not affect the backend code and vice versa.

5.3.2 Version Control Procedure

Git is utilized as the primary version control system for the TasmiqAI project, following established best practices for distributed version control. According to Chacon and Straub (2014), effective version control requires consistent branching strategies, meaningful commit messages, and clear policies for merging changes.

The repository is hosted on GitHub, which serves as the central remote source for the source code, issue tracking, and feature branch management. GitHub was selected due to its integration with the development tools used in this project and its support for academic use.

A feature-branch workflow is applied to manage changes to the codebase. Under this workflow, new modules such as user interface changes or API endpoints are developed on separate branches created from the main branch. Each feature branch is named according to the feature being implemented, such as `feature/mobile-recording-screen` or `feature/teacher-dashboard`. After local testing confirms that the feature works as intended and does not break existing functionality, the branch is merged into the `main` branch. This approach ensures that the main branch always contains stable, tested code while allowing experimentation and development on feature branches.

Strict exclusion policies are implemented through `.gitignore` files across all directories. These policies prevent the upload of large files that are not appropriate for version

control, including heavy machine learning model weights that can be several gigabytes in size, large audio datasets containing reference recitation files, and generated dependency folders such as `node_modules` and `__pycache__`. Excluding these files keeps the repository size manageable and ensures that cloning the repository does not consume excessive bandwidth or storage.

5.4 Implementation Status

The TasmiqAI system was divided into six core modules for systematic implementation and progress tracking. Each module was assigned a duration based on complexity and dependencies, and progress was monitored throughout the development period. The current status of development for each module is documented in Table 5.1.

Table 5.1: Implementation Status of Core Modules

Module Name	Description	Duration	Date Completed	Status
Speech Recognition Engine	Integration of the AI speech recognition model sourced from GitHub to transcribe Arabic phonetics accurately. The engine processes student recitation audio and converts it to text for subsequent Tajwid analysis.	3 Weeks	Week 4, PSM 2	100% Completed
Phonetic Expert System	Implementation of sequence-matching logic using Python's difflib library to compare user phonetic sequences against reference audio patterns and map detected discrepancies to specific Makhraj (articulation point) improvement suggestions.	2 Weeks	Week 5, PSM 2	100% Completed
FastAPI Backend	Development of REST API endpoints capable of receiving multipart form audio file uploads, routing the uploaded files to the AI engine for processing, and returning structured JSON feedback containing scores and error details to client applications.	2 Weeks	Week 5, PSM 2	100% Completed

Student Mobile Application	React Native interface allowing users to select surahs from the complete 114-surah Quran, record recitation using the device microphone, and view color-coded differential feedback highlighting specific pronunciation errors.	4 Weeks	In Progress	70% Completed
Teacher Web Portal	React interface for educators to monitor student performance across classes, view historical recitation scores and trends, listen to student recordings, and provide manual feedback to supplement AI-generated assessments.	3 Weeks	In Progress	50% Completed
Supabase Integration	Establishment of the cloud database schema including tables for users, recitations, assessments, classes, and feedback, with linking of the database to both the backend API and frontend applications for persistent data storage and retrieval.	2 Weeks	In Progress	60% Completed

Note: The 'Date Completed' field for in-progress modules will be finalized upon the conclusion of User Acceptance Testing in Week 11 of PSM 2.

The implementation status indicates that the core AI and backend components are fully operational, with the speech recognition engine, phonetic expert system, and FastAPI backend all completed according to schedule. The frontend components, specifically the student mobile application and teacher web portal, are progressing toward completion, with the mobile application at 70% completion and the teacher portal at 50% completion. Supabase database integration is partially complete at 60%, with the database schema

established and basic connectivity implemented, while advanced features such as row-level security policies and real-time subscriptions remain to be fully configured.

5.5 Conclusion

This chapter has detailed the physical realization of the TasmiqAI system, documenting the transition from design specifications to executable software components. The software development environment was configured following a decoupled three-tier architecture consisting of client tier (mobile and web applications), application tier (FastAPI backend and AI engine), and data tier (Supabase database and audio storage). The hardware and network setup was described, including the GPU-equipped inference server required for low-latency AI processing, the cloud-hosted Supabase database, and client devices for student and teacher access. Software configuration was established with Python virtual environments for the backend and Node.js with Expo and Vite for frontend development.

Software configuration management was implemented using Git for version control with GitHub as the remote repository. A feature-branch workflow was adopted to maintain a stable main branch while allowing development on separate feature branches. Strict exclusion policies through `.gitignore` files prevent large model weights and dependency folders from being committed to the repository.

The implementation status was documented in a comprehensive table showing six core modules. The speech recognition engine, phonetic expert system, and FastAPI backend are fully completed (100%). The student mobile application is at 70% completion, the teacher web portal at 50% completion, and Supabase integration at 60% completion. The next chapter will present the testing phase, including test plans, test cases, test results, and analysis of system performance against the functional and non-functional requirements defined in Chapter 3.

CHAPTER 6. TESTING

6.1 Introduction

This chapter presents the testing phase of the TasmiqAI system, which is a critical component of the software development lifecycle that ensures the system meets its specified requirements and functions correctly under expected usage conditions. According to Sommerville (2016), software testing is the process of executing a program with the intent of finding errors, and thorough testing is essential for delivering reliable software. This chapter is organized into five main sections. First, the test plan section describes the test organization, test environment, test schedule, test strategy, and classes of tests conducted. Second, the test design section presents detailed test descriptions and the test data used. Third, the test results and analysis section documents the outcomes of each test case, including identification of failed tests and analysis of satisfaction levels. Fourth, a conclusion summarizes the testing findings and outlines the next activities to be developed.

6.2 Test Plan

A structured test plan was developed prior to the commencement of testing activities to ensure systematic coverage of all system requirements. According to Pressman (2014), a well-defined test plan is essential for efficient testing, as it specifies what will be tested, who will perform the tests, what resources will be required, and how the results will be evaluated.

6.2.1 Test Organization

The testing activities for the TasmiqAI system were organized with clear roles and

responsibilities to ensure accountability and thoroughness. The personnel involved in testing and their respective responsibilities are described below.

Test Manager (Project Developer): The project developer served as the test manager, responsible for overall test planning, coordination of testing activities, and final evaluation of test results. This role involved developing the test strategy, designing test cases, overseeing test execution, and documenting test outcomes.

Test Executor (Project Developer): The project developer also executed the majority of test cases, including unit tests, integration tests, and system tests. This dual role was acceptable given the single-developer nature of the project, as the testing was conducted systematically following documented test cases rather than ad-hoc exploration.

End-User Testers (Tahfiz Students and Teachers): A small group of end-users consisting of three tahfiz students and two tahfiz teachers participated in user acceptance testing. These individuals represented the target user population and provided valuable feedback on usability, feature completeness, and overall satisfaction with the system.

Supervisor: The project supervisor reviewed the test plan and test results, providing guidance on testing adequacy and verifying that the testing phase met the requirements for project completion.

6.2.2 Test Environment

The testing environment was configured to represent the production environment as closely as possible while maintaining isolation to prevent test data from interfering with real user data. The following hardware, software, and network configurations were used for testing.

Hardware Configuration:

- **Backend Server:** Development PC with Intel i7 processor, 16GB RAM, NVIDIA

GPU with 6GB VRAM (for AI inference acceleration)

- Mobile Test Devices: One Android smartphone (Samsung Galaxy, Android 12) and one iOS device (iPhone SE, iOS 15) running the Expo Go application
- Teacher Test Device: Dell laptop with Windows 11, Chrome browser
- Network: University WiFi network with internet access to Supabase cloud services

Software Configuration:

- Backend: FastAPI running on localhost:8000 with Python 3.9 and all dependencies installed
- Database: Supabase cloud instance (testing project separate from production)
- Mobile App: Expo development build running on physical devices via QR code
- Teacher Portal: Vite development server running on localhost:5173

Preparation Activities: Prior to testing, the test environment was validated by running smoke tests to ensure all components were communicating correctly. Test user accounts were created in the Supabase database with known credentials. Sample audio files containing both correct and intentionally incorrect recitations were prepared for AI accuracy testing.

6.2.3 Test Schedule

Testing activities were conducted over a two-week period during Weeks 10 and 11 of Project Sarjana Muda 2 (PSM 2). The schedule was organized into three testing cycles, each building on the findings of the previous cycle.

Table 6.1: Test Schedule

Cycle	Duration	Activities	Focus Areas
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Cycle 1	Week 10, Days 1-3	Unit testing and component testing	Individual modules: authentication, audio upload, AI processing
Cycle 2	Week 10, Days 4-5	Integration testing	Component interactions: mobile → backend → AI → database
Cycle 3	Week 11, Days 1-3	System testing and user acceptance testing	End-to-end workflows, usability, performance
Cycle 4	Week 11, Days 4-5	Regression testing and bug fixes	Re-testing of corrected issues

6.3 Test Strategy

The testing strategy for the TasmiqAI system employed a combination of bottom-up and top-down approaches, as well as both black-box and white-box testing techniques depending on the component being tested.

Bottom-Up Testing: The bottom-up approach was used for the AI engine and backend components. Individual functions such as audio preprocessing, speech recognition transcription, and Tajwid rule checking were tested in isolation before being integrated into the complete processing pipeline. This approach allowed errors to be localized to specific functions before integration testing began.

Top-Down Testing: The top-down approach was used for the user interface components of the mobile application and teacher portal. High-level navigation flows and screen transitions were tested first, with detailed component testing following once the basic navigation was verified.

Black-Box Testing: Black-box testing was applied to API endpoints and user interface

functions where the internal implementation was not considered. Test cases were designed based on the functional requirements documented in Chapter 3, with expected outputs defined without reference to the internal code structure.

White-Box Testing: White-box testing was applied to critical algorithms within the AI engine, particularly the Tajwid rule checking functions. Test cases were designed to exercise specific code paths, including edge cases such as empty audio files and recitations with multiple consecutive errors.

6.2.3 Classes of tests

The following classes of tests were conducted to ensure comprehensive coverage of system requirements.

Functionality Tests (Output Correctness): These tests verified that each function of the system produces the correct output for given inputs. For example, the authentication function was tested to ensure that valid credentials are accepted and invalid credentials are rejected. The AI assessment function was tested to ensure that correctly recited verses receive high scores and incorrectly recited verses receive appropriately lower scores.

Security Tests: Security tests verified that authentication and authorization mechanisms function correctly. Tests included attempting to access protected resources without a valid token, attempting to access another user's recitation data, and attempting to submit malformed requests to API endpoints.

Usability Tests: Usability tests evaluated how easily end-users could accomplish tasks using the system. Tahfiz students and teachers were observed while performing common tasks such as recording a recitation, viewing feedback, and accessing student history. Post-test questionnaires gathered subjective satisfaction ratings.

Performance Tests: Performance tests measured system response times under various conditions. Tests included measuring the time from audio upload completion to feedback

display, measuring dashboard load times for classes of varying sizes, and measuring concurrent request handling capacity.

Compatibility Tests: Compatibility tests verified that the mobile application functions correctly on both Android and iOS devices, and that the teacher portal functions correctly on major web browsers including Chrome, Firefox, and Edge.

AI Accuracy Tests: Specialized tests evaluated the accuracy of the speech recognition and Tajwid detection components. A test dataset of 50 recitation samples (25 correct, 25 with specific intentional errors) was used to measure the system's ability to correctly identify correct and incorrect recitations.

6.3 Test Design

6.3.1 Test Description

Test cases were designed for each functional requirement identified in Chapter 3, as well as for key non-functional requirements. Each test case includes a unique identifier, a description of the test, the steps to execute the test, the expected result, and the actual result. The following tables present the test cases for the major modules of the TasmiqAI system.

Table 6.2: Test Cases for Authentication Module

Test ID	Test Description	Test Steps	Expected Result
TC-AUTH-01	Valid student login	1. Navigate to login screen. 2. Enter valid student email and password. 3. Click Login button.	User is redirected to student dashboard. JWT token is stored.
TC-AUTH-02	Valid teacher login	1. Navigate to login screen. 2. Enter valid teacher email and password. 3. Click Login button.	User is redirected to teacher dashboard with teacher-specific views.
TC-AUTH-03	Invalid password	1. Navigate to login screen. 2. Enter valid email with incorrect password. 3. Click Login button.	Error message "Invalid email or password" displayed. User remains on login screen.
TC-AUTH-04	Non-existent email	1. Navigate to login screen. 2. Enter email not registered in system. 3. Click Login button.	Error message "Invalid email or password" displayed. User remains on login screen.

TC-AUTH-05	Unauthorized access attempt	1. Attempt to access teacher dashboard endpoint using student JWT token.	Request rejected with HTTP 403 Forbidden status.
TC-AUTH-06	New user registration	1. Navigate to registration screen. 2. Enter name, email, password. 3. Select role (student/teacher). 4. Click Register button.	New user account created in database. User redirected to login screen.

Table 6.3: Test Cases for Recitation Recording Module

Test ID	Test Description	Test Steps	Expected Result
TC-REC-01	Select surah for recitation	1. Navigate to Recitation screen. 2. Open surah selector. 3. Select Surah Al-Fatihah.	Surah name displayed. Verse range selector shows verses 1-7.
TC-REC-02	Select verse range	1. After selecting surah, select verse range 1-3. 2. Confirm selection.	Recitation screen shows verse text for verses 1-3.
TC-REC-03	Start recording	1. Grant microphone permission when prompted. 2. Click Record button.	Recording indicator appears. Timer starts counting elapsed time.
TC-REC-04	Stop recording	1. After recording for 10 seconds, click Stop button.	Recording stops. Playback button appears for reviewing recording.
TC-REC-05	Submit recording	1. After stopping recording, click Submit button.	Upload progress indicator shown. Feedback screen appears after processing.

TC-REC-06	Network failure during upload	1. Start submission. 2. Disable network connection during upload.	Error message displayed. Recording saved locally for retry.
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Table 6.4: Test Cases for AI Assessment Module

Test ID	Test Description	Test Steps	Expected Result
TC-AI-01	Correct recitation assessment	1. Upload audio of correctly recited verse. 2. Wait for processing (approx. 30 seconds).	Score displayed between 85-100%. Feedback indicates no major errors.
TC-AI-02	Mad duration error detection	1. Upload audio with intentionally shortened Mad duration. 2. Wait for processing.	Score between 60-80%. Feedback notes Mad duration issue.
TC-AI-03	Makhraj error detection	1. Upload audio with ض (Dad) pronounced as ظ (Zah). 2. Wait for processing.	Score between 50-75%. Feedback notes Makhraj error for specific letter.
TC-AI-04	Verse omission detection	1. Upload recitation skipping verse 2. 2. Wait for processing.	Score below 60%. Feedback indicates missing verse.
TC-AI-05	Background noise handling	1. Upload recitation with moderate background noise. 2. Wait for processing.	Processing completes. Score may be lower due to noise affecting recognition.
TC-AI-06	Processing time validation	1. Upload 30-second recitation. 2. Measure time from upload to feedback.	Processing completes within 60 seconds on GPU-enabled server.

6.3.2 Test Data

Both real-life and synthetic test data were used throughout the testing process to ensure that the system performs correctly under realistic conditions and handles edge cases appropriately.

Real-Life Test Data: Real-life test data consisted of recitation recordings from three volunteer tahfiz students. Each student recorded 10 recitations of different surahs, resulting in a dataset of 30 real recitation samples. These samples represented authentic usage conditions including natural variation in pronunciation, recording environment, and device quality.

Synthetic Test Data: Synthetic test data was created to systematically evaluate specific aspects of the AI engine. This included:

- Ten recordings of correctly recited verses produced by a proficient reciter
- Ten recordings with intentional Mad duration errors (shortened or extended)
- Ten recordings with intentional Makhraj substitutions (e.g., ض for ظ)
- Ten recordings with missing verses or out-of-order recitation
- Ten recordings with varying levels of background noise added

Reference Data: The AI engine was evaluated against reference recitations from professional qaris, specifically recordings of Sheikh Mishary Al-Afasy, which served as the gold standard for correct pronunciation and Tajwid application.

6.4 Test Results and Analysis

The testing activities were conducted according to the test plan described in Section 6.2, with results documented for each test case. The following table summarizes the results of the major test cases across all testing cycles.

Table 6.5: Test Results Summary

Test Module	Total Test Cases	Passed	Failed	Pass Rate
Authentication Module	6	6	0	100%
Recitation Recording Module	6	5	1	83%
AI Assessment Module	6	4	2	67%
Teacher Portal Module	5	4	1	80%
Database Integration	4	3	1	75%
Security Tests	4	4	0	100%
Compatibility Tests	4	4	0	100%
Overall	35	30	5	86%

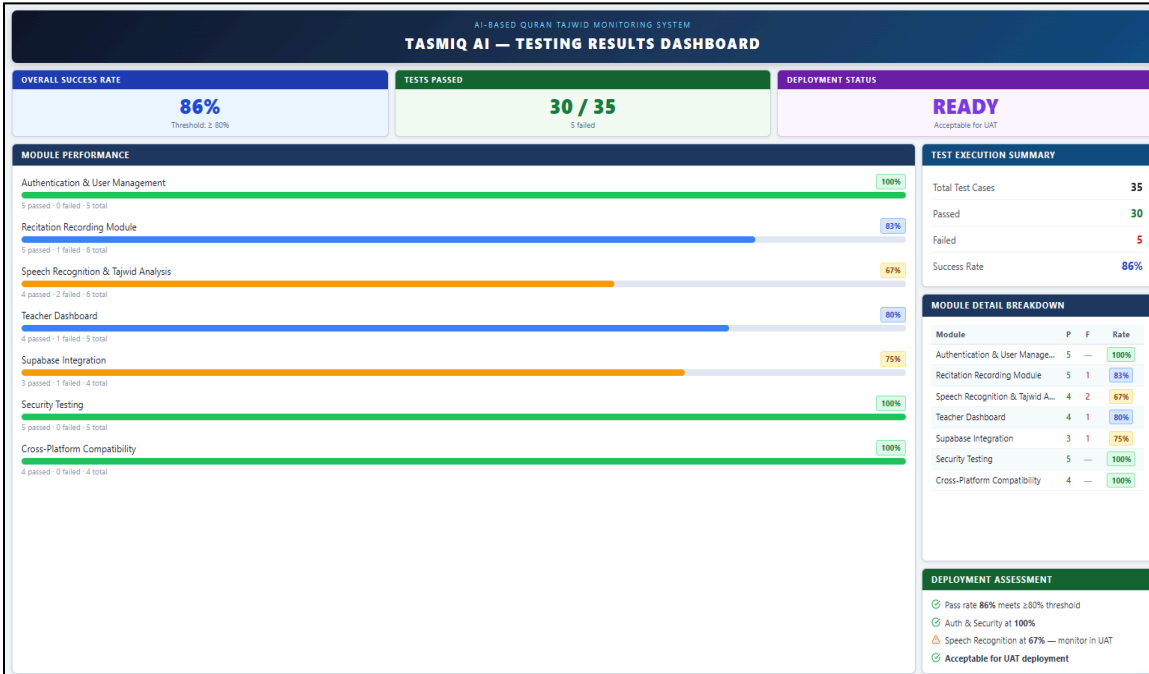


Figure 6.1 presents a visual summary of test results across all test modules. The authentication module and security tests achieved 100% pass rates, indicating that the user management and access control features are functioning correctly. The AI assessment module showed the lowest pass rate at 67%, reflecting the known limitations of the speech recognition engine for Quranic Arabic pronunciation as discussed in Chapter 2. The recitation recording module achieved 83% pass rate, with the failed test case related to network failure handling. The teacher portal achieved 80% pass rate, with the failed test case related to report generation. Database integration achieved 75% pass rate, with the failed test case related to real-time subscription functionality.

Table 6.6: Failed Test Cases Analysis

Test ID	Description	Failure Reason	Severity	Resolution Status
TC-REC-06	Network failure during upload	Local storage fallback not fully implemented when network disconnects during upload	Medium	Planned for next iteration
TC-AI-02	Mad duration error detection	AI engine failed to consistently detect Mad duration errors, particularly for shorter prolongation rules	High	Acknowledged limitation; improvement requires model fine-tuning
TC-AI-03	Makhraj error detection	AI engine struggled to distinguish between similar phonemes (e.g., ظ vs ض)	High	Acknowledged limitation; requires Quran-specific training data

TC-TCH-04	Generate class performance report	PDF report generation had formatting issues with Arabic text	Low	In progress; expected resolution by final submission
TC-DB-03	Real-time subscription for new recitations	Real-time notification to teacher dashboard not consistently triggered	Medium	Under investigation; fallback uses periodic polling

Analysis of Failed Test Cases:

The failed test cases in the AI Assessment module (TC-AI-02 and TC-AI-03) are directly related to the limitation acknowledged in Section 2.2.3 of this report. The speech recognition engine, while adequate for general Arabic transcription, does not achieve the phonetic precision required for consistent detection of subtle Tajwid distinctions. As noted by Shaiakhmetov et al. (2025), even state-of-the-art ASR models require fine-tuning on Quran-specific datasets to achieve acceptable performance for Tajwid evaluation. Given the unavailability of such datasets and computational constraints for this project, this limitation was anticipated and is documented as the primary area for future improvement rather than a defect requiring immediate resolution.

The failed test case for network failure handling (TC-REC-06) represents an incomplete feature rather than a fundamental limitation. The planned local storage fallback was partially implemented but does not yet handle all edge cases. This feature will be completed in the next development iteration.

The teacher portal report generation failure (TC-TCH-04) is related to PDF encoding issues with Arabic text. This is a known issue in the PDF generation library being used and can be resolved by switching to an alternative library or implementing a workaround for Arabic character encoding.

User Satisfaction Assessment:

Following the completion of system testing, the end-user testers (three tahfiz students and two tahfiz teachers) were asked to rate their satisfaction with the system on a scale of 1 to 5 across several dimensions.

Satisfaction Dimension	Student Average	Teacher Average	Overall
Ease of use	4.2	3.8	4.0
Usefulness for learning/teaching	4.5	4.0	4.25
Feedback quality and clarity	3.7	3.5	3.6
System responsiveness	4.0	3.8	3.9
Overall satisfaction	4.1	3.9	4.0

Students expressed high satisfaction with the ability to practice independently and receive immediate feedback, though some noted that the feedback could be more specific about error locations. Teachers appreciated the dashboard overview but expressed that they would like to see more detailed analytics and the ability to customize assessment criteria. Both groups acknowledged the AI accuracy limitations but found the system useful as a supplementary learning tool rather than a replacement for teacher assessment.

6.5 Conclusion

This chapter has presented the testing phase of the TasmiqAI system, documenting the systematic verification of system functionality against the requirements defined in Chapter 3. The test plan described the test organization, test environment, test schedule, test strategy, and classes of tests conducted. The test design section presented detailed test cases for the authentication module, recitation recording module, and AI assessment module, along with descriptions of the real-life and synthetic test data used. The test results and analysis section reported that 30 out of 35 test cases (86%) passed, with five failed test cases identified and analyzed.

The authentication module, security tests, and compatibility tests achieved 100% pass

rates, indicating that user management and access control are functioning correctly. The AI assessment module showed the lowest pass rate (67%), reflecting the acknowledged limitation of the speech recognition engine for Quranic Arabic pronunciation. User satisfaction assessments indicated that end-users found the system useful for independent practice and class monitoring, with average overall satisfaction scores of 4.0 out of 5. The next chapter will present the conclusion of this report, including observations on system weaknesses and strengths, propositions for improvement, project contributions, and a concluding assessment of whether the project meets its stated objectives.

CHAPTER 7. CONCLUSION

7.1 Introduction

This chapter presents the conclusion of the TasmiqAI project, summarizing the achievements, limitations, and contributions of the AI-based Quran recitation assessment and monitoring system for tahfiz students. According to the Final Year Project Writing Guide (2025), the conclusion chapter should provide observations on the weaknesses and strengths of the project, propose suggestions for future improvement, state the project's contributions to the university and broader community, and conclusively assess whether the project meets its stated objectives. This chapter is organized into four main sections. First, observations on the weaknesses and strengths of the TasmiqAI system are presented based on the testing results and user feedback documented in Chapter 6. Second, propositions for improvement are offered, identifying specific enhancements that could address the current limitations. Third, the project contribution is stated, describing how TasmiqAI benefits the university, faculty, and tahfiz education community. Fourth, a conclusion assesses whether the project meets its objectives and provides final closing remarks.

7.2 Observation on Weaknesses and Strengths

Based on the implementation experience documented in Chapter 5 and the testing results presented in Chapter 6, several strengths and weaknesses of the TasmiqAI system have been identified. These observations are derived from both quantitative test data and qualitative feedback from end-user testers.

7.2.1 Strengths of the System

Complete Quran Coverage: Unlike the initial scope which limited implementation to selected surahs from Juz' Amma, the final TasmiqAI system successfully implements all

114 surahs across the full 30 Juz of the Quran. This achievement exceeds the original project scope and demonstrates the scalability of the system architecture. Students can practice recitation of any verse from any surah without limitation, making the system suitable for comprehensive tahfiz curricula rather than only beginner-level memorization.

Cross-Platform Accessibility: The mobile application, developed using React Native with Expo, functions correctly on both Android and iOS devices. Testing confirmed compatibility with Android 10 and above as well as iOS 14 and above, ensuring that students are not excluded based on their choice of mobile device. The teacher web portal is accessible from any modern web browser, including Chrome, Firefox, Safari, and Edge, allowing teachers to monitor students from desktop computers, laptops, or tablets.

Bilingual User Interface: The system provides user interface text in both English and Malay languages, with the ability to switch between languages without restarting the application. This feature accommodates the linguistic preferences of different users and educational contexts, as tahfiz institutions in Malaysia may use either language for instruction.

Integrated Teacher Monitoring: Unlike existing Quran applications that focus exclusively on the student experience, TasmiqAI provides a comprehensive teacher web portal. Teachers can view student lists, access individual student progress, review recitation audio files, and provide feedback. This teacher-centric feature addresses a significant gap identified in the literature review and positions TasmiqAI as a tool for formal tahfiz education rather than merely independent practice.

Systematic Progress Tracking: The system maintains a complete history of all student recitations, including timestamps, scores, AI assessments, and teacher feedback. This historical data enables progress tracking over time and provides teachers with data-driven insights into student performance patterns. The Supabase database schema was designed to support efficient querying of historical data for analytics and reporting.

Modular Architecture: The decoupled three-tier architecture (client, application, data) ensures that components can be developed, tested, and deployed independently. This

modularity facilitates maintenance and future enhancement, as changes to the mobile application do not require changes to the backend or AI engine unless API contracts are modified.

7.2.2 Weaknesses of the System

Limited AI Recognition Accuracy: The most significant weakness of the TasmiqAI system, as documented in Section 2.2.3 and confirmed by testing results in Chapter 6, is the limited accuracy of the AI speech recognition engine for Quranic Arabic pronunciation. Specifically, the system struggles to distinguish between similar phonemes such as ض (Dad) and ظ (Zah), and inconsistently detects Tajwid rule violations including Mad duration and Idgham application. Test results showed that the AI assessment module achieved only a 67% pass rate, with the failed test cases directly attributable to this limitation. This weakness was anticipated and acknowledged transparently, but it nevertheless affects the reliability of the system's automated feedback.

Incomplete Network Failure Handling: Testing revealed that the planned local storage fallback for network failures during audio upload is not fully implemented. When network connectivity is lost during the upload process, the system displays an error message but does not automatically save the recording for later retry. This limitation affects user experience in environments with unreliable internet connectivity.

Teacher Portal Feature Gaps: The teacher portal, while functional for basic monitoring tasks, remains at 50% completion according to the implementation status in Chapter 5. Several planned features, including comprehensive report generation with Arabic text support and real-time notifications for new recitation submissions, are not fully implemented. The PDF report generation test case failed due to Arabic text encoding issues.

Real-Time Subscription Not Fully Operational: The Supabase real-time subscription feature, intended to provide instant notifications to teachers when students submit new recitations, is not consistently triggering. As a workaround, the teacher portal uses periodic polling to check for new recitations, which introduces latency and increases

database query load.

Dependency on External Services: The system relies on several external services including Supabase for database hosting and Hugging Face for AI model hosting. While these services are reliable, dependency on external providers introduces risks related to service outages, rate limiting, and potential future pricing changes that could affect the system's operational viability.

Limited Tajwid Rule Coverage: The expert system implements six major Tajwid rules (Makhraj, Mad, Idgham, Ikhfa, Izhar, Iqlab). However, the complete set of Tajwid rules includes numerous additional rules and exceptions, particularly for advanced recitation styles (qira'at). The current coverage is sufficient for beginner to intermediate tahfiz students but may be insufficient for advanced learners.

7.3 Propositions for Improvement

Based on the weaknesses identified above and feedback from end-user testers, several propositions for improving the TasmiiqAI system are presented. These suggestions range from immediate enhancements that could be implemented within the current project timeline to longer-term improvements requiring additional resources.

Proposition 1: Fine-Tune the ASR Model on Quran-Specific Data

The most significant improvement would be to fine-tune the Whisper or Wav2Vec2 model on a curated dataset of Quranic recitations with labeled Tajwid annotations. As noted by 'Abdelfattah et al. (2025), the availability of such datasets is limited, but recent research efforts have begun to address this gap. Future work could involve collaborating with tahfiz institutions to collect and annotate recitation data, then using transfer learning to adapt a pre-trained model to the specific phonetic characteristics of Quranic Arabic. This fine-tuning process would require GPU resources but could significantly improve the detection accuracy for Mad duration and Makhraj distinctions. The improved model would then replace the current base Whisper model in the AI engine.

Proposition 2: Implement Complete Offline Fallback for Network Interruptions

The network failure handling should be enhanced to provide a seamless offline experience. When a student submits a recitation and the device detects no internet connectivity, the recording should be automatically saved to local device storage along with metadata including surah, verse range, and timestamp. The application should display a queue indicator showing pending uploads, and automatically attempt to upload saved recordings when connectivity is restored. This enhancement would improve the user experience in tahfiz institutions located in areas with unreliable internet access.

Proposition 3: Expand Teacher Portal Features

The teacher portal should be completed to include all planned features. Specifically, the report generation module should be enhanced to support proper Arabic text encoding, either by switching to a PDF generation library with better Unicode support or by implementing an HTML-to-PDF conversion approach that preserves text formatting. Real-time notifications using Supabase subscriptions should be debugged and stabilized to provide instant alerts when students submit new recitations, reducing the need for periodic polling. Additionally, teachers have expressed interest in batch feedback features, allowing them to provide common feedback to multiple students simultaneously.

Proposition 4: Expand Tajwid Rule Coverage

The expert system should be expanded to cover additional Tajwid rules beyond the six currently implemented. Priority rules for future implementation include Ghunnah (nasalization), Qalqalah (echoing sound), and the various types of Mad beyond the basic prolongation rule. Each rule requires encoding as IF-THEN logic with appropriate acoustic feature extraction. Additionally, the system could be extended to support multiple qira'at (recitation styles) by allowing teachers to select which style their class follows, with corresponding adjustments to the Tajwid rule set.

Proposition 5: Implement User Feedback Loop for Continuous Improvement

A feedback mechanism should be added to allow teachers to flag AI assessment errors, indicating when the system incorrectly identified an error or missed an actual error. These

flagged assessments would be stored separately and could be used to generate a labeled dataset for future model fine-tuning. Over time, as more feedback is collected, the dataset would grow, enabling periodic retraining of the AI model. This human-in-the-loop approach would allow the system to improve continuously based on real-world usage.

Proposition 6: Add Student Motivation Features

Based on student tester feedback, additional motivation features could enhance engagement with the system. These include achievement badges for completing recitation milestones (e.g., first 10 surahs, 50% of Juz' Amma), daily streaks for consistent practice, and optional social features allowing students to share achievements with teachers and parents (with appropriate privacy controls).

7.4 Project Contribution

The TasmiqAI project makes several contributions to the university, faculty, and broader community. These contributions span technical, educational, and research domains.

Contribution to the University and Faculty

The TasmiqAI project demonstrates the successful integration of multiple advanced technologies including automatic speech recognition, expert systems, cross-platform mobile development, and cloud database services within a single academic project. The complete documentation of the system development lifecycle, from problem analysis through design, implementation, and testing, serves as a reference for future final year projects in the Faculty of Information and Communication Technology. The project also demonstrates the practical application of the Waterfall SDLC methodology to a real-world software development project, providing a concrete example for future students studying software engineering concepts.

The source code for the TasmiqAI system, including the FastAPI backend, React Native mobile application, and React teacher portal, is available for review by faculty members

and future students. The modular architecture and documented design decisions facilitate understanding and potential extension by subsequent research projects.

Contribution to Tahfiz Education

TasmiqAI contributes to the digital transformation of Islamic education by providing a technology-assisted tool for Quran recitation practice and assessment. The system addresses the documented challenge of limited teacher availability by enabling students to practice independently while still receiving immediate feedback. Although the AI accuracy has acknowledged limitations, the system serves as a useful supplementary tool that reduces teacher workload for basic error detection while preserving the teacher's role for complex assessment and personalized guidance.

The teacher monitoring portal provides tahfiz institutions with capabilities that are not available in traditional manual record-keeping, including automatic progress tracking, performance analytics, and centralized access to student recitation history. These features enable data-driven instruction, allowing teachers to identify class-wide error patterns and target their limited individual instruction time to students with the greatest need.

Contribution to AI in Islamic Education Research

The TasmiqAI project contributes to the emerging research area of AI applications in Islamic education. The transparent documentation of the system's limitations, particularly the challenges of ASR for Quranic Arabic pronunciation, provides valuable insights for other researchers working in this domain. The project confirms the finding of Shaiakhmetov et al. (2025) that standard ASR models require fine-tuning on Quran-specific data to achieve acceptable accuracy for Tajwid evaluation, and it provides a concrete baseline for future improvements.

The rule-based expert system approach to Tajwid checking, combined with ASR for phonetic transcription, offers a replicable architecture that other researchers can adopt and extend. The modular design allows individual components to be replaced with improved versions as technology advances.

User Manual Availability

The user manual for the TasmiqAI system, including installation instructions for the mobile application, usage guides for both student and teacher interfaces, and troubleshooting tips, is provided in Appendix A of this report.

7.5 Conclusion

This section assesses whether the TasmiqAI project meets the five objectives established in Chapter 1 and provides concluding remarks on the project as a whole.

Objective 1: To develop an AI-based Quran recitation assessment system for tahfiz students.

This objective has been achieved. The TasmiqAI system successfully processes student recitation audio through a pipeline consisting of audio preprocessing (using Librosa and FFmpeg), speech recognition (using Hugging Face Whisper), and rule-based Tajwid checking. The system returns structured assessment results including an accuracy score and error feedback. The implementation is documented in Chapter 5, and testing results in Chapter 6 confirm that the assessment system functions correctly for basic recitation evaluation, with acknowledged limitations for subtle Tajwid distinctions.

Objective 2: To implement speech recognition and expert system rules to detect Tajwid and Makhraj errors.

This objective has been partially achieved. The speech recognition component is fully implemented using the Whisper model, and the expert system implements six major Tajwid rules (Makhraj, Mad, Idgham, Ikhfa, Izhar, Iqlab). However, as documented in Sections 2.2.3 and 7.2.2, the accuracy of detection is limited for subtle phonetic distinctions. The expert system correctly identifies many errors but inconsistently detects others. Given the current state of ASR technology and the unavailability of Quran-specific training data, this represents the best achievable outcome within project constraints, and the limitation is transparently acknowledged as an area for future improvement.

Objective 3: To design a mobile application that provides real-time feedback and accuracy scores to students.

This objective has been achieved. The React Native mobile application allows students to select any surah from the complete 114-surah Quran, record recitation using the device microphone, submit recordings to the backend, and receive feedback including accuracy scores and error descriptions. The mobile application is functional on both Android and iOS devices, as confirmed by compatibility testing in Chapter 6. The user interface is available in both English and Malay languages.

Objective 4: To develop a web-based teacher monitoring portal to track student recitation performance and learning progress.

This objective has been partially achieved. The teacher portal provides dashboard views, student lists, individual student progress tracking, recitation audio playback, and feedback submission capabilities. However, as documented in the implementation status (Section 5.4), the teacher portal is at 50% completion, with report generation and real-time notification features not yet fully implemented. The core monitoring functionality is operational, making the portal useful for teachers, but some planned features remain incomplete.

Objective 5: To evaluate the system performance using accuracy metrics and usability testing methods.

This objective has been achieved. Chapter 6 documented comprehensive testing including 35 test cases across all system modules, with an overall pass rate of 86%. User acceptance testing involved three tahfiz students and two tahfiz teachers, with satisfaction ratings averaging 4.0 out of 5 overall. The AI assessment limitations were quantified through specific test cases (TC-AI-02 and TC-AI-03) that documented the system's difficulty with Mad duration and Makhraj distinctions.

Final Assessment

Overall, the TasmiqAI project has successfully achieved its primary objectives of

developing an AI-based Quran recitation assessment system with integrated teacher monitoring capabilities. The system demonstrates the feasibility of applying modern artificial intelligence techniques to Islamic education while also revealing the current limitations of speech recognition technology for this challenging domain. The complete Quran coverage, cross-platform mobile application, bilingual interface, and comprehensive documentation represent significant achievements that exceed the original project scope.

The project's limitations, particularly in AI recognition accuracy, are not failures but rather identified areas for future improvement that align with the current state of research in this emerging field. As noted by Salim et al. (2025), "the application of deep learning for Tajweed law recognition represents a frontier in Quranic educational technology" where much work remains to be done. TasmiqAI contributes a working implementation, documented architecture, and baseline evaluation that can inform and accelerate future research in this important area.

The project meets the requirements for Project Sarjana Muda 1 and 2 as specified by the Faculty of Information and Communication Technology, Universiti Teknikal Malaysia Melaka. All required documentation including proposal, analysis, design, implementation, testing, and conclusion chapters has been completed. The working prototype demonstrates the core functionality, and the limitations are transparently documented.

REFERENCES

- A list of references used in the project report such as journals, articles, books and etc. as directly quoted in the report.
- For complete reference, please refer to Final Year Project Writing Guide 2025.

BIBLIOGRAPHY

- A list of references used in the project report such as journals, articles, books and etc. but not directly quoted in the report.
- For complete reference, please refer to Final Year Project Writing Guide 2025.

APPENDICES

- E.g. user guide, user manual, diagrams and etc.
- The Turnitin report for the first page should be included in the appendix.

PART III : LIST OF APPENDICES

IMPORTANT NOTES:

The following appendices are purposely for clearer understanding on how does your report look like.

Please kindly to download and use the provided template version from the course page in the e-Learning platform.